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Case-Based Reasoning: Foundational Issues, Methodological Variations, and System Approaches

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Case-based reasoning is a recent approach to problem solving and learning that has got a lot of attention over the last few years. Originating in the US, the basic idea and underlying theories have spread to other continents, and
we are now within a period of highly active research in case-based reasoning in Europe, as well. This paper gives an overview of the foundational issues related to case-based reasoning, describes some of the leading methodological approaches within the field, and exemplifies the current state through pointers to some systems. Initially, a general framework is defined, to which the subsequent descriptions and discussions will refer. The framework is influenced by recent methodologies for knowledge level descriptions of intelligent systems. The methods for case retrieval, reuse, solution testing, and learning are summarized, and their actual realization is discussed in the light of a few example systems that represent different CBR approaches. We also discuss the role of case-based methods as one type of reasoning and learning method within an integrated system architecture.

1. Introduction

Over the last few years, case-based reasoning (CBR) has grown from a rather specific and iso-lated research area to a field of widespread interest. Activities are rapidly growing - as seen by the increased rate of research papers, availability of commercial products, and also reports on applications in regular use. In Europe, researchers and application developers recently met at the First European Workshop on Case-based reasoning, which took place in Germany, November 1993. It gath-ered around 120 people and more than 80 papers on scientific and application-oriented research were presented.

1.1. Background and motivation.

Case-based reasoning is a problem solving paradigm that in many respects is fundamentally differ-ent from other major AI approaches. Instead of re-lying solely on general knowledge of a problem domain, or making associations along generalized relationships between problem descriptors and conclusions, CBR is able to utilize the *specific* knowledge of previously experienced, concrete problem situations (cases). A new problem is solved by finding a similar past case, and reusing it in the new problem situation. A second important difference is that CBR also is an approach to incre-mental, sustained learning, since a new experience is retained each time a problem has been solved, making it immediately available for future prob-lems.

This paper presents an overview of the field, in terms of its underlying foundation, its current state-of-the-art, and future trends. The description of CBR principles, methods, and systems is made within a general analytic scheme. Other authors have recently given overviews of case-based rea-soning (Ch. 1 in [51], Introductory section of [18], [36,61]). Our overview differs in four major ways from these accounts: First, we initially specify a general descriptive framework to which the subse-quent method descriptions will refer. Second, we put a strong emphasis on the methodological issues of case-based reasoning, and less on a discussion of suitable application types and on the advantages of CBR over rule-based systems. (This has been taken very well care of in the documents cited above). Third, we strive to maintain a neutral view of existing CBR ap-

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基于案例的推理：基础性问题，方法学差异与系统化方法

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基于案例的推理是一种新兴的问题求解与学习方法，近年来受到了广泛关注。在过去几年中。起源于美国，其基本思想并且相关理论已传播至其他大洲。和我们目前正处于一项高度活跃的研究阶段。在欧洲，基于案例的推理也同样得到了应用。本文概述了与案例相关的基本问题。基于推理的方法，描述了一些主要的故障诊断方法。该领域内的逻辑方法，并通过引用若干系统来阐明当前的研究现状。起初，定义了一个通用框架，后续的描述与讨论将予以引用。该框架受到近期知识表示方法的影响。对智能系统的描述。案例检索、重用、解法验证及学习的方法已概述。已予定型，其具体实现将在下文中予以讨论。几个代表不同CBR方法的示例系统。我们还讨论了基于案例的方法所发挥的作用。OODS作为一种推理与学习方法，其内部一种集成的系统架构。

1. 引言

在过去几年中，基于案例的推理（CBR）已从一种相对特定且孤立的方法发展成为一种更为广泛的方法研究领域发展为一个广受关注的学科方向。活动正在迅速增长——正如……所显示的那样研究论文发表数量的增加、商业产品的普及，以及关于应用的报告在日常使用中。在欧洲，研究人员和应用应用程序开发者最近在首届欧洲会议上会面。欧洲基于案例的推理研讨会于1993年11月在德国举行。它汇集了约120人及80余篇科学论文科学性与应用导向的研究已提出。

1.1. Background and motivation.

基于案例的推理是一种问题求解范式，在许多方面与人工智能的其他主要方法存在根本性差异。它并不依赖于……仅凭对某一问题领域的通用知识，o通过在问题描述符与结论之间建立广义关系来进行关联，案例推理 i能够运用以往所经历的具体问题情境（案例）的*specific* 知识。通过寻找相似的过往案例来解决新问题。并在新的问题情境中加以重用。第二个重要区别在于，案例推理也是一种方法从而实现渐进式、持续性的学习，因为每当一个问题得到解决时，新的经验便会被保留下来，并立即可用于未来的应用。问题。

本文从其理论基础和当前发展状况两个方面对该领域进行了综述。最先进的技术及未来发展趋势。对基于案例推理的原理、方法和系统的描述是在……范围内进行的。一般的分析框架。其他作者近期也对基于案例的推理进行了综述（第1章 i[51]，[18]的绪论部分，[36,61]）。欧概述在四个方面与这些有所不同账户：首先，我们初步定义了一个通用的描述性框架，其后续方法描述将予以引用。其次，我们提出高度重视案例的方法论问题基于推理，而较少关注适用的应用类型以及案例推理相较于其他方法的优势。基于规则的系统。（这一点已被广泛接受）在上述引用的文献中予以关注）。第三，我们力求对现有的基于案例的推理方法保持中立的立场。

proaches, unbiased by a particu-lar 'school'[1]. And finally, we include results from the European CBR arena, which unfortunately have been missing in American CBR reports.

What is case-based reasoning? Basically: To solve a new problem by remembering a previous similar situation and by reusing information and knowledge of that situation. Let us illustrate this by looking at some typical problem solving situations:

- A physician - after having examined a particu-lar patient in his office - gets a reminding to a patient that he treated two weeks ago. Assuming that the reminding was caused by a similarity of important symptoms (and not the patient's hair-color, say), the physician uses the diagnosis and treatment of the previous patient to determine the disease and treatment for the patient in front of him.
- A drilling engineer, who have experienced two dramatic blow out situations, is quickly re-minded of one of these situations (or both) when the combination of critical measurements matches those of a blow out case. In particular, he may get a reminding to a mistake he made during a previous blow-out, and use this to avoid repeating the error once again.
- A financial consultant working on a difficult credit decision task, uses a reminding to a pre-vious case, which involved a company in similar trouble as the current one, to recommend that the loan application should be refused.

1.2. Case-based problem solving.

As the above examples indicate, reasoning by reusing past cases is a powerful and frequently app-lied way to solve problems for humans. This claim is also supported by results from cognitive psychological research. Part of the foundation for the case-based approach, is its psychological plau-sibility. Several studies have given empirical evi-dence for the dominating role of specific, pre-viously experienced situations (what we call cases) in human problem solving (e.g. [53]). Schank [54] developed a theory of learning and reminding based on retaining of experience in a dynamic, evolving memory[2] structure. Anderson [6] has shown that people use past cases as models when learning to solve problems, particularly in early learning. Other results (e.g. by W.B. Rouse [75]) indicate that the use of past cases is a predominant problem solving method among experts as well. Studies of problem solving by analogy (e.g. [16,22]) also shows the frequent use of past experience in solving new and different problems. Case-based reasoning and analogy are sometimes used as synonyms (e.g. in [16]). Case-based reasoning can be considered a form of *intra-domain analogy*. However, as will be discussed later, the main body of analogical research [14,23,30] has a differ-ent focus, namely analogies across domains.

In CBR terminology, a *case* usually denotes a *problem situation*. A previously experienced situation, which has been captured and learned in such way that it can be reused in the solving of future problems, is referred to as a past case, previous case, stored case, or retained case. Correspond-ingly, a new case or unsolved case is the description of a new problem to be solved. Case-based reasoning is - in effect - a cyclic and integrated process of solving a problem, learning from this experience, solving a new problem, etc.

Note that the term problem solving is used here in a wide sense, coherent with common practice within the area of knowledge-based systems in general. This means that problem solving is not necessarily the finding of a concrete solution to an application problem, it may be any problem put forth by the user. For example, to justify or criti-cize a solution proposed by the user, to interpret a problem situation, to generate a set of possible so-lutions, or generate expectations in observable data are also problem solving situations.

1.3. Learning in Case-based Reasoning.

A very important feature of case-based reason-ing is its coupling to learning. The driving force behind case-based methods has to a large extent come from the machine learning community, and case-based reasoning is also regarded as a subfield of machine

[1]Our own experience from active CBR research over the last 5 years started out from different backgrounds and motivations, and we may have developed different views to some of the major issues involved. We will give examples of our respective priorities and concerns related to CBR research as part of the discussion about future trends towards the end of the paper.

[2]The term 'memory' is often used to refer to the storage structure that holds the existing cases, i.e. to the case base. A memory, thus, refers to what is remembered from previous experiences. Correspondingly, a reminding is a pointer structure to some part of memory.

方法，不受某一特定“学派”的偏见影响[1]。最后，我们还纳入了来自欧洲案例推理竞技场的结果，而这些结果此前一直缺失。美国CBR报告。

什么是基于案例的推理？基本上：为了解决通过记忆先前的类似问题而引发了一个新问题通过利用当前情境，并复用该情境中的信息与知识。让我们通过考察某些情况来对此加以说明。典型的问题解决情境：

- 一位医师——在对某一特定情况进行检查后——患者在他的办公室——收到一条关于患者的提醒他两周前诊治的那例。假设表明该提醒是由某种相似性引起的。重要的症状（而非患者的头发）

例如，在诊断过程中，医生会参考前一位患者的诊断与治疗方案，以此来确定当前患者的疾病及其治疗方案。——一位曾经历过两次严重井喷事故的钻井工程师，很快便被再次提醒……

当关键测量值的组合满足其中一种（或两种）情形时，那些属于井喷事故的情况。特别是，他可能会获得提醒他之前犯下的一个错误爆破，并利用这一点以避免重复出现该错误。

再一次。——一位正在处理一项复杂的信贷决策任务的财务顾问，通过提醒自己先前的一个案例来辅助判断。

其中涉及一家与当前公司面临类似困境的企业，以建议……

贷款申请应予拒绝。

1.2. Case-based problem solving.

正如上述示例所示，通过复用过往案例进行推理是一种强大且广泛应用的方法。为人类解决问题的一种方式。这一主张也得到了认知心理学研究结果的支持研究。案例库是其基础的一部分。该方法的心理合理性是其一大优势。Several研究已为特定的、先前的经验所发挥的主导作用提供了实证证据。人类问题解决中的各种情境（我们称之为案例）（例如[53]）。Schank [54] 提出了一个理论……基于保持的学习与提醒在动态且不断演化的记忆结构中获得的经验。Anderson [6] 已经证明，人们会利用过往的案例来…模型在学习解决问题时，尤其是在早期学习阶段。其他研究结果（例如由 W.B. Rous 所得）[75] 表明，借鉴过往案例是专家群体中一种占主导地位的问题解决方法。同样地，关于类比问题解决的研究（例如[16,22] 也表明，在解决新的、不同的问题时，人们常常会借鉴以往的经验。案例推理与类比有时也被用于同义词（例如在[16]中）。基于案例的推理可以被视为一种*intra-domain analogy*。然而，正如后文将要讨论的，类比研究的主体部分[14,23,30]具有不同的焦点，即跨领域的类比。

在基于案例的推理术语中，*case* 通常指代一个*problem situation*。即一个先前经历过的、以某种方式被记录并学习的情境，使得……它可以在未来问题的求解中被重复使用，被称为过往案例、先前案例、存储案例或保留案例。相应地，新案例或未解案例是指待解决的新问题的描述。基于案例的推理实际上是一个循环且集成的问题求解过程。问题、从该经验中学习、解决新问题等。

请注意，此处“问题解决”一词的使用范围较广，与该领域内的普遍做法保持一致。知识系统领域的一般情况。这意味着问题求解并不一定是……为某一应用问题找到具体解决方案，该问题可以是用户提出的任何问题。例如，用于论证或批评用户提出的解决方案，以及对问题情境进行解释。生成一组可能的解决方案，或生成对可观测数据的期望同样属于问题求解情境。

1.3. Learning in Case-based Reasoning.

基于案例的推理的一个非常重要的特征是其与学习的耦合是其背后的驱动力。基于案例的方法在很大程度上源自机器学习领域，而案例库推理也被视为机器学习的一个子领域。

[1]Our own experience from active CBR research over the last 5 years started out from different backgrounds and motivations, and we may have developed different views to some of the major issues involved. We will give examples of our respective priorities and concerns related to CBR research as part of the discussion about future trends towards the end of the paper.

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learning[3]. Thus, the notion of case-based reasoning does not only denote a particular reasoning method, irrespective of how the cases are acquired, it also denotes a machine learning paradigm that enables sustained learning by updating the case base after a problem has been solved. Learning in CBR occurs as a natural by-product of problem solving. When a problem is successfully solved, the experience is retained in order to solve similar problems in the future. When an attempt to solve a problem fails, the reason for the failure is identified and remembered in order to avoid the same mistake in the future.

Case-based reasoning favours learning from experience, since it is usually easier to learn by retaining a concrete problem solving experience than to generalize from it. Still, effective learning in CBR requires a well worked out set of methods in order to extract relevant knowledge from the expe-rience, integrate a case into an existing knowledge structure, and index the case for later matching with similar cases.

1.4. Combining cases with other knowledge.

By examining theoretical and experimental results from cognitive psychology, it seems clear that human problem solving and learning in general are processes that involve the representation and utili-zation of several types of knowledge, and the com-bination of several reasoning methods. If cognitive plausibility is a guiding principle, an architecture for intelligence where the reuse of cases is at the centre, should also incorporate other and more general types of knowledge in one form or another. This is an issue of current concern in CBR research [67].

The rest of this paper is structured as follows: The next section gives a brief historical overview of the CBR field. This is followed by a grouping of CBR methods into a set of characteristic types, and a presentation of the descriptive framework which will be used throughout the paper to discuss CBR methods. Sections 4 to 8 discuss representation is-sues and methods related to the four main tasks of case-based reasoning, respectively. In section 9 we look at CBR in relation to integrated architectures and multistrategy problem solving and learning. This is followed by a short description of some fielded applications, and a few words about CBR development tools. The conclusion part briefly summarizes the paper, and point out some possi-ble trends.

[3]The learning approach of case-based reasoning is sometimes referred to as case-based learning. This term is sometimes also used synonymous with example-based learning, and may therefore point to classical induction and other generalization-driven learning methods. Hence, we will here use the term case-based reasoning both for the problem solving and learning part, and explicitly state which part we talk about whenever necessary.

2. History of the CBR field

The roots of case-based reasoning in AI are found in the works of Roger Schank on dynamic memory and the central role that a reminding of earlier situations (episodes, cases) and situation patterns (scripts, MOPs) have in problem solving and learning [54]. Other trails into the CBR field have come from the study of analogical reasoning [22], and - further back - from theories of concept formation, problem solving and experiential learning within philosophy and psychology (e.g. [62,69,72]). For example, Wittgenstein observed that ‘natural concepts’, i.e., concepts that are part of the natural world - such as bird, orange, chair, car, etc. - are polymorphic. That is, their instances may be categorized in a variety of ways, and it is not possible to come up with a useful classical definition in terms of a set of necessary and sufficient features for such concepts. An answer to this problem is to represent a concept extensionally, defined by its set of instances - or cases.

The first system that might be called a case-based reasoner was the CYRUS system, developed by Janet Kolodner [33, 34], at Yale University (Schank's group). CYRUS was based on Schank's dynamic memory model and MOP theory of prob-lem solving and learning [54]. It was basically a question-answering system with knowledge of the various travels and meetings of former US Secre-tary of State Cyrus Vance. The case memory model developed for this system has later served as basis for several other case-based reasoning systems (in-cluding MEDIATOR [59], PERSUADER [68], CHEF [24], JULIA [27], CASEY [38]).

Another basis for CBR, and another set of models, were developed by Bruce Porter and his group [48] at the University of Texas, Austin. They initially addressed the machine learning problem of concept learning for classification tasks. This lead to the development of the PROTOS system [9], which emphasized integrating general domain knowledge and specific case knowledge into a unified representation structure. The combination of cases with general domain

学习[3]。因此，基于案例的推理这一概念不仅指代一种特定的推理方法。，无论病例是如何获得的，它也o表示一种使……成为可能的机器学习范式s通过在问题求解后更新案例库来实现持续学习。在基于案例的推理中，学习过程表现为a问题解决的自然副产品。当a问题已成功解决，经验i s以备将来解决类似问题之用。当尝试解决问题失败时，e故障原因被识别并记录在案，以避免未来再次发生同样的错误。

基于案例的推理有利于从现有-经验，因为通常通过保留具体的解决问题经验来学习比从中进行概括更容易。然而，在基于案例的推理中实现有效学习，仍需一套经过精心设计的方法，以便从经验中提取出相关知识。-因此，将案例整合到现有的知识结构中，并为后续匹配建立索引。克与类似案例。

1.4. Combining cases with other knowledge.

通过考察认知心理学的理论与实验研究结果，似乎可以明确地得出结论：人类n问题解决与学习在总体上是过程。s涉及表示与利用的f多种类型的知识，以及其组合f多种推理方法。若认知合理性为a指导原则是，一种以案例复用为核心的智能架构，同时还应纳入其他更为通用的知识类型。e以某种形式存在。这是一个当前的问题。t在案例推理研究中备受关注[67]。

本文的其余部分安排如下：Th e下一节将简要回顾CBR领域的历史发展。随后，我们将CBR方法归纳为若干具有代表性的类型，并a描述性框架的呈现，其将所有将在整篇论文中用于讨论基于案例的推理方法。。第4至8节讨论了与案例库四大主要任务相关的表示问题及方法。d推理，分别进行。在第9节中，我们探讨基于案例的推理与集成架构以及多策略问题求解和学习之间的关系。随后是a若干已部署应用的简要介绍，以及a关于CBR开发工具的几点说明。结论部分简要总结了本文，一个d指出一些可能的趋势。

2. CBR领域的研究历程

基于案例的推理在人工智能领域的根源可以追溯到在罗杰·香克关于动态记忆的研究中，d早期情境的提醒所发挥的中心作用s（事件、病例）与情境模式（脚本），MOPs）在问题解决与学习中具有重要作用[54]].其他通往CBR领域的研究路径则源于……e对类比推理的研究[22]，以及——更早之前-源自哲学与心理学领域中关于概念形成、问题解决及体验式学习的理论-心理学（例如[62,69,72]）。例如，维特根斯坦n观察到“自然概念”，即那些e自然界的一部分——例如鸟、橙子、茶r,汽车等——是多态的。也就是说，它们的实例s可能以多种方式加以分类，而且它并非t有可能提出一个基于一组必要且充分特征的、具有实用性的经典定义。s对于此类概念，一种解决方案是采用外延方式来表示概念，即通过其集合来加以定义。f实例——或案例。

最早可被称为基于案例的推理系统的是由简开发的CYRUS系统。t科洛德纳[33, 34]，在耶鲁大学（尚克）s组）。CYRUS 基于沙克的动态c记忆模型与问题解决的MOP理论克以及学习[54]。它本质上是一个具备多种知识的问答系统。s美国前国务卿的出访与会晤e赛勒斯·万斯。所开发的案例记忆模型r该系统后来已成为多个其他系统的基础。r基于案例的推理系统（包括 MEDIATOR [59]、PERSUADER [68]、CHEF [24]、JULIA [27] 和 CASEY [38]）。

CBR的另一理论基础以及另一组模型是由布鲁斯·波特及其研究团队[48]所提出的。t德克萨斯大学奥斯汀分校。他们最初着手解决概念学习这一机器学习问题。t用于分类任务的学习。这导致了开发l-PROTOS系统的开发[9]，该系统强调整合通用领域知识与特定c将案例知识整合为统一的表示结构。案例与通用领域知识的结合n

[3]The learning approach of case-based reasoning is sometimes referred to as case-based learning. This term is sometimes also used synonymous with example-based learning, and may therefore point to classical induction and other generalization-driven learning methods. Hence, we will here use the term case-based reasoning both for the problem solving and learning part, and explicitly state which part we talk about whenever necessary.

knowledge was pushed further in GREBE [12], an application in the domain of law. Another early significant contribution to CBR was the work by Edwina Rissland and her group at the University of Massachusetts, Amhearst. With sev-eral law scientists in the group, they were interested in the role of precedence reasoning in legal judgements [52]. Cases (precedents) are here not used to produce a single answer, but to interpret a situation in court, and to produce and assess argu-ments for both parties. This resulted in the HYPO system [10], and later the combined case-based and rule-based system CABARET [60]. Phyllis Koton at MIT studied the use of case-based reason-ing to optimize performance in an existing knowledge based system, where the domain (heart fail-ure) was described by a deep, causal model. This resulted in the CASEY system [38], in which case-based and deep model-based reasoning was com-bined.

In Europe, research on CBR was taken up a little later than in the US. The CBR work seems to have been stronger coupled to expert systems develop-ment and knowledge acquisition research than in the US. Among the earliest results was the work on CBR for complex technical diagnosis within the MOLTKE system, done by Michael Richter to-gether with Klaus Dieter Althoff and others at the University of Kaiserslautern [4]. This lead to the PATDEX system [50], with Stefan Wess as the main developer, and later to several other systems and methods [5]. At IIIA in Blanes, Enric Plaza and Ramon López de Mántaras developed a case-based learning apprentice system for medical diag-nosis [46], and Beatrice Lopez investigated the use of case-based methods for strategy-level reasoning [39]. In Aberdeen, Derek Sleeman's group studied the use of cases for knowledge base refinement. An early result was the REFINER system, developed by Sunil Sharma [57]. Another result is the IULIAN system for theory revision by Ruediger Oehlman [44]. At the University of Trondheim, Agnar Aamodt and colleagues at Sintef studied the learning aspect of CBR in the context of knowl-edge acquisition in general, and knowledge main-tenance in particular. For problem solving, the combined use of cases and general domain knowl-edge was focused [1]. This lead to the development of the CREEK system and integration framework [2], and to continued work on knowledge-intensive case-based reasoning. On the cognitive science side, early work was done on analogical reasoning by Mark Keane, at Trinity College, Dublin, [29], a group that has developed into a strong environment for this type of CBR. In Gerhard Strube's group at the University of Freiburg, the role of episodic knowledge in cognitive models was investigated in the EVENTS project [66], which lead to the group"s current research profile of cognitive science and CBR.

Currently, the CBR activities in the United States as well as in Europe are spreading out (see, e.g. [8,19,20,28], and the rapidly growing number of papers on CBR in almost any AI journal). Germany seems to have taken a leading position in terms of number of active researchers, and several groups of significant size and activity level have been established recently. From Japan and other Asian countries, there are also activity points, for example in India [71]. In Japan, the interest is to a large ex-ent focused on the parallel computation approach to CBR [32].

3. Fundamentals of case-based reasoning methods

Central tasks that all case-based reasoning meth-ods have to deal with are to identify the current problem situation, find a past case similar to the new one, use that case to suggest a solution to the current problem, evaluate the proposed solution, and update the system by learning from this experi-ence. How this is done, what part of the process is focused, what type of problems drives the methods, etc. varies considerably, however. Below is an at-tempt to classify CBR methods into types with roughly similar properties in this respect.

3.1. Main types of CBR methods.

The CBR paradigm covers a range of different methods for organizing, retrieving, utilizing and indexing the knowledge retained in past cases. Cases may be kept as concrete experiences, or a set of similar cases may form a generalized case. Cases may be stored as separate knowledge units, or splitted up into subunits and distributed within the knowledge structure. Cases may be indexed by a prefixed or open vocabulary, and within a flat or hierarchical index structure. The solution from a previous case may be directly applied to the present problem, or modified according to differ-ences between the two cases. The matching of cases, adaptation of solutions, and learning from an

相关知识在GREBE [12]中得到了进一步推进，在法律领域的应用。另一个早期对案例推理（CBR）作出重要贡献的工作是埃德温娜·里斯兰及其团队在……大学马萨诸塞州，阿默斯特。与数名法律科学家在该组中，它们相互在法律推理中，优先性推理的作用已被研究。判断[52]。在此，判例并未被使用。不是为了得出单一答案，而是为了对某一情境进行解读。在法庭上，并提出和评估论据双方。这促成了HYPO系统[10]的诞生。随后，基于案例与规则相结合的系统CABARET[60]问世。菲莉丝·科托麻省理工学院研究了基于案例的推理的应用。在现有基于知识的系统中优化性能，其中领域为（心力衰竭）由一个深层的因果模型所描述。这导致了……CASEY 系统 [38]，其中基于案例的方法与深度基于模型的推理被结合使用。

在欧洲，关于案例推理的研究起步较美国稍晚。与美国相比，欧洲的案例推理研究似乎更紧密地结合了专家系统开发与知识获取方面的研究。

最早的成果之一是关于基于案例推理的研究。由迈克尔·里希特与克劳共同完成的MOLTKE系统内的复杂技术诊断迪特尔·阿尔托夫及凯泽斯劳滕大学的其他研究人员[4]。这促成了PATDEX系统的开发[50]，其中斯特凡·韦斯为主要开发者，随后又衍生出若干其他系统。方法[5]。在布拉内斯的IIIA，恩里克·普拉萨和拉蒙·洛佩斯·德·曼塔拉斯开发了一个案例基于学习的医学诊断学徒系统诺西斯[46]，而贝亚特丽斯·洛佩斯则研究了基于案例的方法在战略层推理中的应用[39]。在阿伯丁，德里克·斯利曼的团队研究了……的使用情况。用于知识库精化的案例。一项早期的研究是Sunil Sharma开发的REFINER系统[57]。另一个成果是由Ruediger Oehlman提出的用于理论修正的IULIAN系统[44]。在特隆赫姆大学，阿格纳尔·奥莫特SINTEF的同事们在知识获取的背景下研究了案例推理的学习方面。以及尤其是知识维护。对于问题求解，案例的综合运用……通用领域知识受到了关注[1]。这一研究对……的发展CREEK 系统与集成框架 [2]，并继续开展知识密集型案例的研究基于推理。在认知科学方面另一方面，早期的研究主要集中在类比推理方面。都柏林三一学院的马克·基恩[29]，一个群体已经发展成为一种适用于此类基于案例推理的强大环境。在格哈德·斯特鲁贝的研究组中，弗赖堡大学，情景知识的作用在认知模型中的应用已在EVENTS项目中得到研究[66]，这促成了该研究团队当前在认知科学与案例推理领域的研究方向。

目前，美国及欧洲的案例推理活动正在逐步扩展（参见，例如[8,19,20,28]，以及迅速增长的……几乎任何人工智能期刊上关于案例推理的论文）。德语似乎在……方面已处于领先地位活跃研究人员的数量，以及若干个研究小组近期已确认了一些规模和活动水平均较为显著的活动点。此外，来自日本及其他亚洲国家也存在若干活动点，例如印度[71]。在日本，人们的兴趣主要集中在……专注于基于并行计算的案例推理方法[32]。

3. 基于案例的推理基础方法

所有基于案例的推理方法的核心任务是运维人员需要处理的问题是识别当前在遇到问题情境时，首先查找与当前问题相似的过往案例，利用该案例为当前问题提供解决方案，对所提出的方案进行评估，并通过此次经验更新系统。如何……已完成，过程的重点在于哪一部分，属于哪种类型问题的多样性推动了方法的发展等。然而，显著地。以下是一个尝试将案例推理方法大致划分为若干类型，这些类型在某种程度这方面的性质。

3.1. Main types of CBR methods.

案例推理范式涵盖了用于组织、检索和利用的一系列不同方法。对以往案例中所保留的知识进行索引。案例可以作为具体的经验保存，或者以一组形式保存。类似案例可能构成一个广义案例。案例可能可作为独立的知识单元进行存储，也可被分解为子单元并分布于知识结构之中。案例可通过预定义或开放式词汇进行索引，并可在扁平式或层次化索引体系中组织。结构。先前案例中的解决方案可以直接应用于当前问题，或根据差异进行相应调整。在两种情形之间。通过案例匹配、解决方案的适配以及从……中学习

experience may be guided and supported by a deep model of general domain knowledge, by more shallow and compiled knowledge, or be based on an apparent, syntactic similarity only. CBR meth-ods may be purely self-contained and automatic, or they may interact heavily with the user for support and guidance of its choices. Some CBR method assume a rather large amount of widely distributed cases in its case base, while others are based on a more limited set of typical ones. Past cases may be retrieved and evaluated sequentially or in parallel.

Actually, "case-based reasoning" is just one of a set of terms used to refer to systems of this kind. This has lead to some confusions, particularly since case-based reasoning is a term used both as a gen-eric term for several types of more specific ap-proaches, as well as for one such approach. To some extent, this can also be said for analogy rea-soning. An attempt of a clarification, although not resolving the confusions, of the terms related to case-based reasoning are given below.

Exemplar-based reasoning. The term is derived from a classification of different views to concept definition into "the classical view", "the proba-bilistic view", and "the exemplar view" (see [62]). In the exemplar view, a concept is defined *exten-sionally*, as the set of its exemplars. CBR methods that address the learning of concept definitions (i.e. the problem addressed by most of the research in machine learning), are sometimes referred to as exemplar-based. Examples are early papers by Kibler and Aha [31], and Bareiss and Porter [48]. In this approach, solving a problem is a *classifica-tion task*, i.e., finding the right class for the unclas-sified exemplar. The class of the most similar past case becomes the solution to the classification problem. The set of classes constitutes the set of possible solutions. Modification of a solution found is therefore outside the scope of this method.

Instance-based reasoning. This is a specializa-tion of exemplar-based reasoning into a highly *syn-tactic* CBR-approach. To compensate for lack of guidance from general background knowledge, a relatively large number of instances are needed in order to close in on a concept definition. The representation of the instances are usually simple (e.g. feature vectors), since the major focus is on studying *automated learning* with no user in the loop. Instance-based reasoning labels recent work by Kibler and Aha and colleagues [7], and serves to distinguish their methods from more knowledge-intensive exemplar-based approaches (e.g. Protos' methods). Basically, this is a non-generalization approach to the concept learning problem ad-dressed by classical, inductive machine learning methods.

Memory-based reasoning. This approach em-phasizes a collection of cases as a *large memory*, and reasoning as a process of accessing and searching in this memory. Memory organization and access is a focus of the case-based methods. The utilization of *parallel processing* techniques is a characteristic of these methods, and distinguishes this approach from the others. The access and stor-age methods may rely on purely syntactic criteria, as in the MBR-Talk system [63], or they may at-tempt to utilize general do-main knowledge, as the work done in Japan on massive parallel memories [32].

Case-based reasoning. Although case-based reasoning is used as a generic term in this paper, the *typical* case-based reasoning methods have some characteristics that distinguish them from the other approaches listed here. First, a typical case is usually assumed to have a certain degree of rich-ness of information contained in it, and a certain complexity with respect to its internal organization. That is, a feature vector holding some values and a corresponding class is not what we would call a typical case description. What we refer to as typi-cal case-based methods also has another charac-teristic property: They are able to modify, or adapt, a re-trieved solution when applied in a different problem solving context. Paradigmatic case-based methods also utilizes general background knowl-edge - although its richness, degree of explicit re-resentation, and role within the CBR processes varies. Core methods of typical CBR systems bor-row a lot from cognitive psychology theories.

Analogy-based reasoning. This term is some-times used, as a synonym to case-based reasoning, to describe the typical case-based approach just described [70]. However, it is also often used to characterize methods that solve new problems based on past cases from a *different domain*, while typical case-based methods focus on indexing and matching strategies for single-domain cases. Re-search on analogy reasoning is therefore a subfield concerned with mechanisms for identification and utilization of cross-domain analogies [23, 30]. The major focus of study has been on the *reuse* of a past case, what is called the mapping problem: Finding a way to transfer, or map, the solution of an identified analogue (called source or base) to the present problem (called target).

经验可以在深度学习的指导下并得到其支持。 p
通用领域知识模型，由更为浅层且经过编译的知识构成，或基于一种显性的 ，
仅基于句法相似性。案例推理方法可能是纯粹的 y
自成一体且自动运行，或者它们也可能相互交互。 t
在支持与指导方面，该系统与用户高度耦合。 s
选择。一些基于案例的推理方法假设一个相当大的e
其案例库中广泛分布的案例数量较多，而另一些则基于更为有限的一组典型案例。 l
过去的案例可以按顺序或并行地被检索和评估。

实际上，“基于案例的推理”只是其中的一种··等·用于指代此类系统的术语。这已 s
导致一些混淆，尤其是因为基于案例的推理这一术语既被用作一个通用概念，又 r
几种更为具体的处理方法，以及 s
对于其中一种方法而言，某种程度上，这也适用于类比推理。尽管这一尝试并未消除相关困惑，但…… f
与基于案例的推理相关的术语如下: n
如下。

Exemplar-based reasoning. 该术语源于对不同观点的一种分类。 t
定义为“经典观点”、“概率论 c
“视图”以及“范例视图”（参见[62]）。在范例视图中，一个概念被定义为由其所有范例所构成的集合。基于案例的推理方法通过处理这些 e
概念定义的学习（即机器学习领域大多数研究所关注的问题） e
学习），有时被称为实例 -
基于。例如，早期的论文由Kibler和Ah撰写。 a
[31]，以及Bareiss和Porter [48]。在该方法中 ，
解决问题是一个*classifica-tion task*的过程，即为未分类的样本找到正确的类别。该类 s
与当前问题最为相似的过往案例将作为该分类问题的解决方案。类别的集合构成了所有可能的解集。因此，对已找到解的修改超出了该方法的适用范围。

Instance-based reasoning. 这是将基于实例的推理专门化为一种高度形式化的CBR方法。为了弥补缺乏来自一般背景知识的指导，采用了相对较大的 e
需要一定数量的实例才能收敛到o。 n
一个概念的定义。实例的表示通常较为简单（例如，特征向量）。 ，
由于主要关注的是在无人参与的情况下对*automated*和*learning*进行研究，基于实例的推理将Kibler和Aha近期的工作归类为 d
同事[7]，并用于区分他们的方法 s
来自更多知识密集型的示例驱动的方法（例如，Protos 的方法）。基本上，这是 a
针对经典归纳式机器学习方法所处理的概念学习问题的非泛化性方法。

Memory-based reasoning. 该方法将一组案例视为*large memory*，并将推理视为在该记忆库中进行访问与检索的过程。记忆的组织与访问是 a
案例法的重点。该方法的运用 f
*parallel processing*技术是这些方法的特征，并将该方法与其他方法区分开来。访问和存储方法可能依赖于 y
仅依据纯句法标准，如MBR-Talk系统[63]中所采用的那样，或者它们也可能尝试利用通用的…… -
主要知识，正如在日本开展的关于大规模的研究工作并行存储器[32]。

Case-based reasoning. 尽管在本文中，基于案例的推理被用作一个通用术语 ，
基于案例的推理方法具有一些使其有别于此处列出的其他方法的特征。首先，一个典型的案例通常是 y
被认为具有一定程度的丰富性 f
其中包含的信息以及一定的复杂性 y
就其内部组织而言。也就是说， a
包含若干数值的特征向量及其对应的类别，并非我们所称的 a
典型病例描述。我们所指的典型 -
基于案例的计算方法还具有另一项重要特性：它们能够修改或适应一个…… -
在不同的问题求解情境中应用时所获得的解决方案。范例式的基于案例的方法也 o
利用一般的背景知识——尽管其丰富程度、明确表述的程度以及作用 e
在案例推理过程中，其具体实现方式各不相同。典型案例推理系统的核心方法大量借鉴了认知科学的e
相关理论。

Analogy-based reasoning. 该术语有时被用作基于案例的推理的同义词，t o
描述典型的基于案例的方法 t
被[70]所描述。然而，它也常用于表征那些基于*different domain*中的过往案例来解决新问题的方法，而典型的基于案例的方法则侧重于索引…… d
针对单域情形的匹配策略。重新搜索 h
因此，类比推理是一个相关的子领域。 d
并配备用于识别与利用跨领域类比的机制[23, 30]。研究的主要焦点一直在于对某一历史案例的*reuse*，即所谓的映射问题：寻找一种方法以实现迁移 ，
或映射，已识别模拟问题的解（称为 d
从源或基础到当前问题（称为目标）。

Throughout the paper we will continue to use the term case-based reasoning in the generic sense, although our examples, elaborations, and discussions will lean towards CBR in the more typical sense. The fact that a system is described as an example of some other approach, does not exclude it from being a typical CBR system as well. To the degree that more special examples of, e.g., instance-based, memory-based, or analogy-based methods will be discussed, this will be stated explicitly.

3.2. A descriptive framework.

Our framework for describing CBR methods and systems has two main parts:

- A process model of the CBR cycle
- A task-method structure for case-based reasoning

The two models are complementary and represent two views on case-based reasoning. The first is a dynamic model that identifies the main subprocesses of a CBR cycle, their interdependencies and products. The second is a task-oriented view, where a task decomposition and related problem solving methods are described. The framework will be used in subsequent parts to identify and discuss important problem areas of CBR, and means of dealing with them.

3.3. The CBR cycle

At the highest level of generality, a general CBR cycle may be described by the following four processes[4]:

1. RETRIEVE the most similar case or cases
2. REUSE the information and knowledge in that case to solve the problem
3. REVISE the proposed solution
4. RETAIN the parts of this experience likely to be useful for future problem solving

A new problem is solved by *retrieving* one or more previously experienced cases, *reusing* the case in one way or another, *revising* the solution based on reusing a previous case, and *retaining* the new experience by incorporating it into the exist-ing knowledge-base (case-base). The four proc-esses each involve a number of more specific steps, which will be described in the task model. In fFg. 1, this cycle is illustrated.

An initial description of a problem (top of Fig. 1) defines a *new case*. This new case is used to RETRIEVE a case from the collection of *previous cases*. The *retrieved case* is combined with the new case - through REUSE - into a *solved case*, i.e. a proposed solution to the initial problem. Through the REVISE process this solution is tested for success, e.g. by being applied to the real world environment or evaluated by a teacher, and repaired if failed. During RETAIN, useful experience is retained for future reuse, and the case base is updated by a new *learned case*, or by modification of some existing cases.

As indicated in the figure, general knowledge usually plays a part in this cycle, by supporting the CBR processes. This support may range from very weak (or none) to very strong, depending on the type of CBR method. By general knowledge we here mean general domain-dependent knowledge, as opposed to specific knowledge embodied by cases. For example, in diagnosing a patient by re-trieving and reusing the case of a previous patient, a model of anatomy together with causal relation-ships between pathological states may constitute the general knowledge used by a CBR system. A set of rules may have the same role.

3.4. A hierarchy of CBR tasks

The process view just described was chosen in order to emphasize on CBR as a cycle of sequential steps. To further decompose and describe the four top-level steps, we switch to a task-oriented view, where each step, or subprocess, is viewed as a task that the CBR reasoner has to achieve. While a process-oriented view enables a global, external

[4]As a mnemonic, try "the four REs".

在整篇论文中，我们将继续使用 th e 术语“基于案例的推理”在广义上指代…… - 尽管我们的示例、阐述和讨论将更多地倾向于以较为典型的意义上理解基于案例的推理。 e 事实上，一个系统被描述为某种示例 e 另一种方法并不排除其成为 a 典型的案例推理系统亦是如此。在某种程度上，更多特殊的示例，例如基于实例的、记忆 - 基于规则的方法或基于类比的方法将被讨论。 , 这一点将予以明确说明。

3.2. A descriptive framework.

我们用于描述CBR方法的框架 d 该系统主要由两部分组成:

- CBR循环的过程模型 - 基于案例推理的任务-方法结构

这两个模型互为补充，并且代表 t 关于基于案例的推理的两种观点。第一种是 a 能够识别主要子过程的动态模型 f 一个基于案例的推理（CBR）周期，及其相互依赖关系和产出。第二部分则从任务导向的视角出发，描述了任务分解及相关的问题求解方法。该框架将在后续章节中用于识别并讨论基于案例的推理中的重要问题领域及其应对策略。 h 他们。

3.3. The CBR cycle

在最高层次的抽象概括下，一个通用的案例推理循环可由以下四个过程描述: - 过程[4]:

1. 检索最相似的一个或多个案例；2. 复用该案例 中的信息与知识。 e
3. 解决问题；4. 修改所提出的解决方案；5. 保留本次经验中可能对未来问题解决具有参考价值的部分。

一个新的问题通过*retrieving* 一个或多个先前的经验案例得以解决，*reusing*该案例在某种程度上与当前问题相关，*revising*其解决方案基于 在重用先前案例时，以及通过将其纳入现有知识库（案例库）来构建新的经验 *retaining*。这四个过程每 h 涉及若干更为具体的步骤，这些步骤将 e 在任务模型中有所描述。在图1中，该循环i s 已图示。

问题的初始描述（图1顶部） ） 定义了一个*new case*。该新案例用于从*previous cases*的案例库中检索一个案例。通过重用过程，将*retrieved case*与该新案例相结合，形成一个*solved case*，即针对初始问题的初步解决方案。随后，通过修订过程对该解决方案的成功性进行验证，例如通过…… 克 应用于真实环境或被评估 b y 一名教师，并在失败时进行修复。在“保留”阶段 ，将有用的经验予以保留，以供未来重复使用，而e 该案例 知识库通过新增*learned case*或对部分现有案例进行修改而得到更新。

如图所示，一般知识通常在该循环中发挥作用 ，通过支持…… e 案例推理过程。这种支持可能涵盖从验证到 y 从较弱（或无）到非常强，具体取决于类型。 e CBR方法的。在本文中，我们所指的一般知识是指n 与……相反，通用领域相关知识 o 由案例所体现的特定知识。例如 ， 在通过检索并复用既往患者的病案来诊断患者时，解剖学模型以及病理状态之间的因果关系可构成案例推理系统所使用的通用知识；一组规则亦可发挥相同的作用。

3.4. A hierarchy of CBR tasks

上述过程视图的选取旨在强调案例推理作为一种序列化循环的过程。 l 步骤。为进一步分解并描述这四个顶层 - 在各个层级步骤中，我们转而采用任务导向的视角， 其中每个步骤或子过程都被视为一项任务，其 e 案例推理推理机必须实现。而面向过程的 d 视图 enables 一个全局的、外部的 l

[4]As a mnemonic, try "the four REs"。

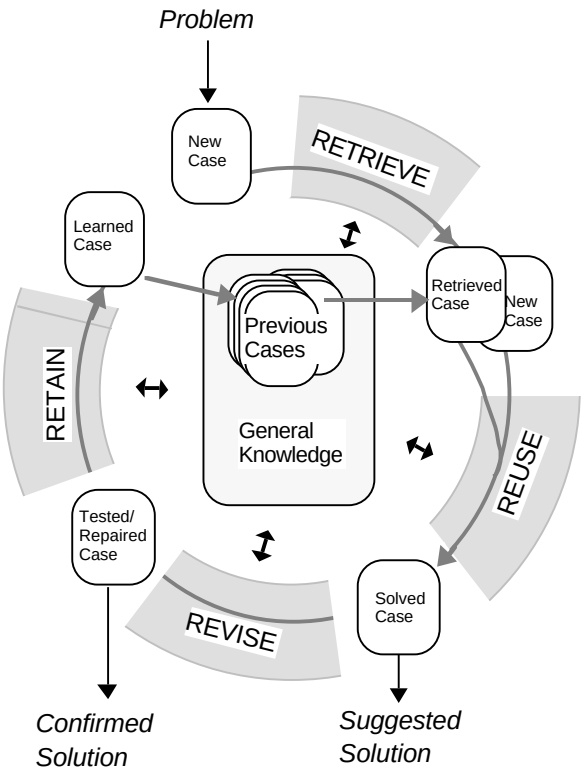


Fig. 1. The CBR Cycle

view to what is happening, a task oriented view is suitable for describing the detailed mechanisms from the perspective of the CBR reasoner itself. This is coherent with a task-oriented view of knowledge level modeling [73]. At the knowledge level, a system is viewed as an agent which has goals, and means to achieve its goals. A system de-scription can be made from three perspectives: Tasks, methods and domain knowledge models. Tasks are set up by the goals of the system, and a task is performed by applying one or more meth-ods. For a method to be able to accomplish a task, it needs knowledge about the general application do-main as well as information about the current prob-lem and its context. Our framework and analysis approach is strongly influenced by knowledge level modeling methods, particularly the Components of Expertise methodology [64.65].

The task-method structure we will refer to in subsequent parts of the paper is shown in Fig. 2. Tasks have node names in bold letters, while meth-ods are written in italics. The links between task nodes (plain lines) are *task decompositions*, i.e., part-of relations, where the direction of the rela-tionship is downwards. The top-level task is prob-lem solving and learning from experience and the method to accomplish the task is *case-based rea-soning* (indicated in a special way by a stippled ar-row). This splits the top-level task into the four major CBR tasks corresponding to the four proc-esses of Fig.1, retrieve, reuse, revise, and retain. All the four tasks are necessary in order to perform the top-level task. The retrieve task is, in turn, partitioned in the same manner (by a retrieval method) into the tasks identify (relevant descriptors), search (to find a set of past cases), initial match (the relevant descriptors to past cases), and select (the most similar case).

All task partitions in the figure are complete, i.e. the set of subtasks of a task are intended to be sufficient to accomplish the task, at this level of description. The figure does not show any control structure over the subtasks, although a rough sequencing of them is indicated by having put ear-lier subtasks higher up on the page than those that follow (for a particular set of subtasks). The actual control is specified as part of the problem solving method. The relation between tasks and methods (stippled lines) identify alternative methods ap-plicable for solving a task. A method specifies the algorithm that identifies and controls the execution of subtasks, and accesses and utilizes the knowl-edge and information needed to do this. The meth-ods shown are high level method classes, from which one or more specific methods should be cho-sen. The method set as shown is incomplete, i.e. one of the methods indicated may be sufficient to solve the task, several methods may be combined, or there may be other methods that can do the job. The methods shown in the figure are task decom-position and control methods. At the bottom level of the task hierarchy (not shown), a task is solved directly, i.e. by what may be referred to as task execution methods.

3.5. CBR Problem Areas

As for AI in general, there are no universal CBR methods suitable for every domain of application. The challenge in CBR as elsewhere is to come up with methods that are suited for problem solving and learning in particular subject domains and for par-ticular application environments. In line with the task model just shown, core problems addressed

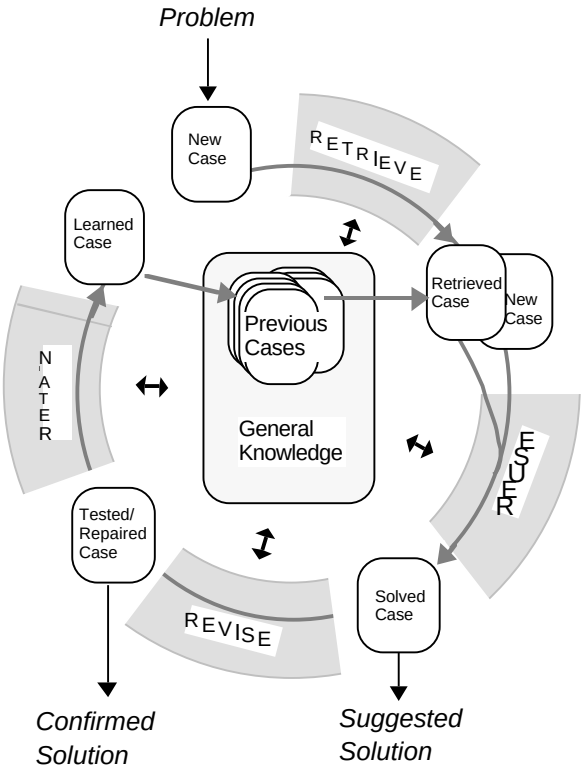


图1. CBR循环

查看正在发生的情况，以任务为导向的视图 i s
适用于从CBR推理机自身的角度描述详细机制。这
与一种面向任务的知识层次观相一致。 l
建模[73]。在知识层面，一个系统i s
被视为具有目标和手段的主体，且 o
实现其目标。可以对系统进行描述。 e
从三个视角出发：任务、方法和领域知识模型。任
务由……的目标所设定。 e
系统，并且通过应用一个o来执行一项任务。 r
更多方法。为了使某种方法能够实现 h
一项任务，需要关于该属的知识。 l
应用领域以及有关……的信息 e
当前问题及其背景。我们的框架 d
分析方法在很大程度上受到知识的影响。 e
层次建模方法，尤其是专业知识要素法[64,65]。

我们在论文后续部分将提及的任务-方法结构如
图2所示。 。
任务的节点名称以粗体显示，而方法 s
以斜体书写。任务节点之间的链接 s
（实线）表示 *task decompositions*，即部分—整体
关系，其中关系的方向为 i s
向下。顶层任务是问题求解。

并且，从经验中学习以及完成该任务的方法是基于
案例的推理（以虚线箭头以特殊方式加以标注）。
这 s
将顶层任务分解为四个主要的CBR任务。 s
对应于图1中的四个过程：检索、重用、修改和保
留。为了完成顶层任务，这四项任务均不可或缺。
e
检索任务又被以相同的方式进行划分。 r
（通过一种检索方法）将任务分解为：识别（相关
描述符）、搜索（以查找一组历史案例）、初步匹
配（将相关描述符与历史案例进行匹配），以及选
择（最相似的案例）。

图中所有任务划分均为完备的，即在该层次上
，某一任务的子任务集合被设计为足以完成该任务
。 -
说明。该图未显示任何对照。 l
在子任务之上构建结构，尽管通过较早地安排某些
子任务已暗示了其大致的顺序。 r
页面上位于前面的子任务比后续的子任务具有更高
的优先级（对于某一组特定的子任务而言）。实际s
的控制
作为问题解决方法的一部分而被明确规定。 e
任务与方法之间的关系（虚线）标识了可用于解决
问题的替代方法。 a
任务。方法指定了用于识别和控制子任务执行的算
法，并访问和利用完成该任务所需的知识与信息。
所展示的方法为高层次的方法类，从中可派生出一
个或多个具体方法。 s
应为“cho-sen”。所设定的方法不完整，即所列出
的某一种方法可能足以解决该任务，也可能需要将
多种方法加以组合，或者还可能存在其他可行的方
法。 o
该任务。图中所示的方法为任务分解与控制方法。
在任务层次结构的最底层（未予展示），一个任务
被求解。 d
直接地，即通过所谓的事后分析技术（TAS） k
执行方法。

3.5. CBR Problem Areas

就人工智能而言，尚不存在适用于所有应用领
域的通用案例推理方法。 e
案例推理与其他领域一样所面临的挑战，在于提出
适用于特定学科领域问题求解与学习的方法，以及
用于…… -
特定的应用环境。与刚刚展示的任务模型相一致，
所解决的核心问题为

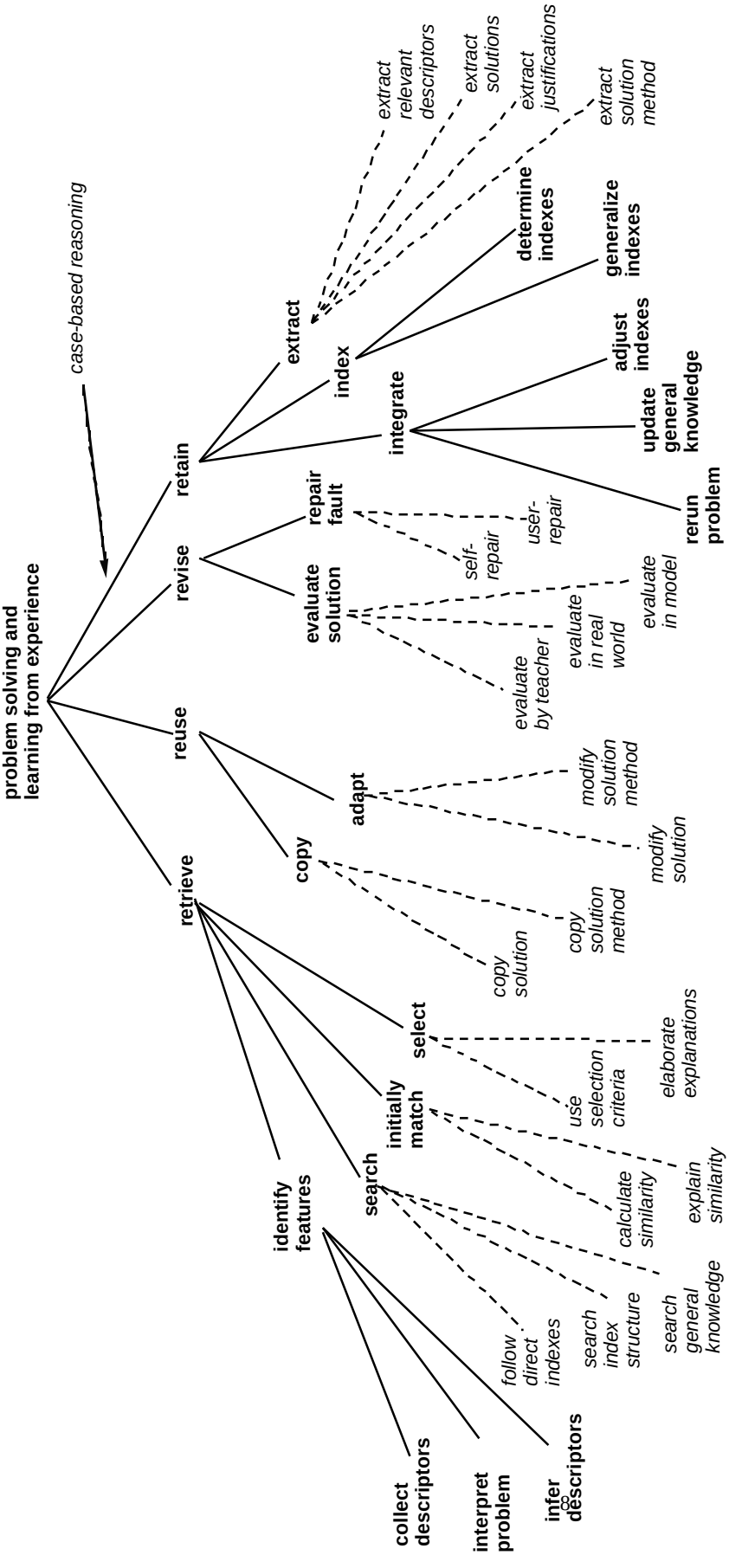
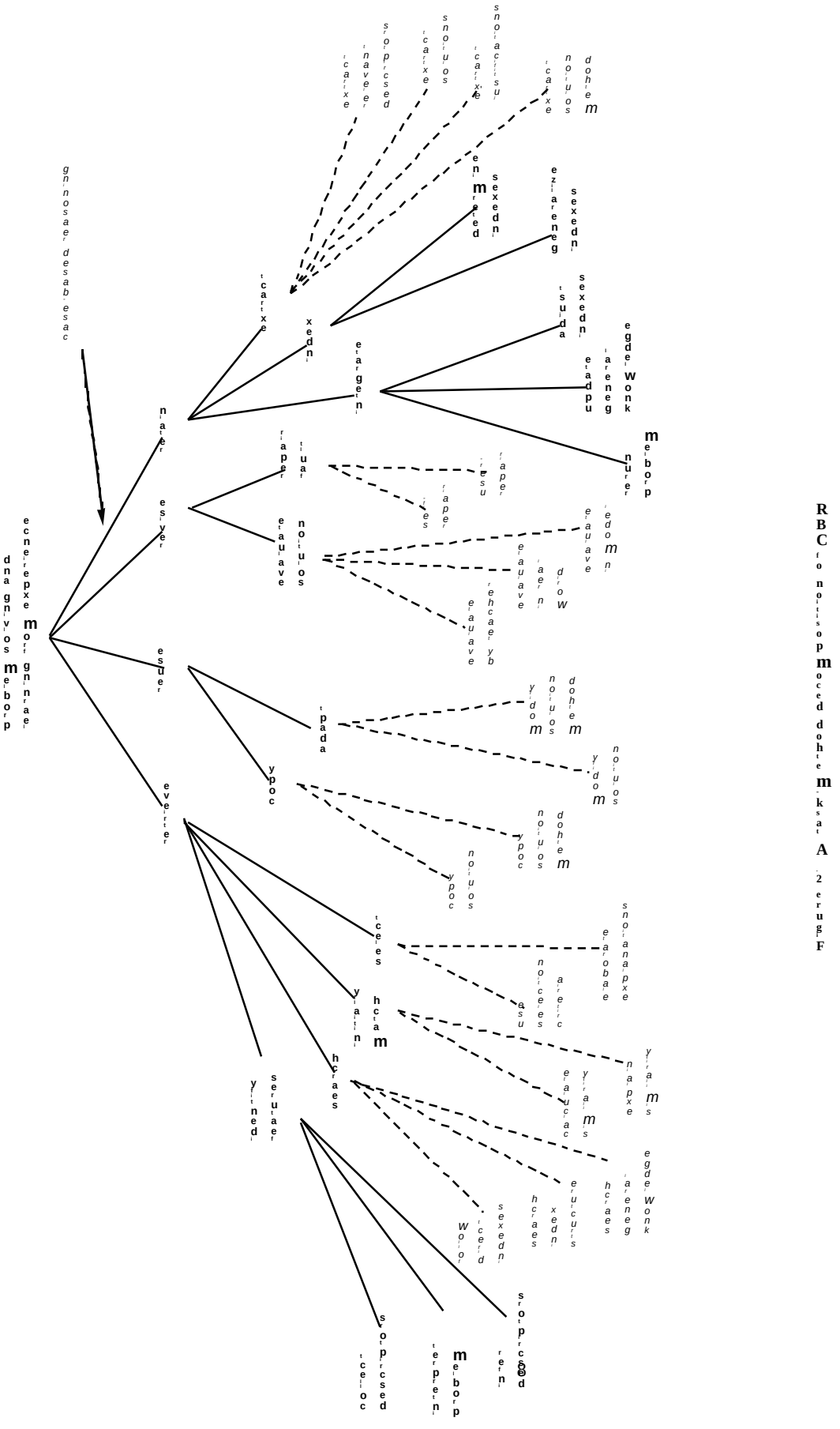


Figure 2. A task-method decomposition of CBR


RBC to notisopmocedd ohtemkcatA .2 erugiF

by CBR research can be grouped into five areas. A set of coherent solutions to these problems constitutes a CBR method:

- *Knowledge representation*
- *Retrieval methods*
- *Reuse methods*
- *Revise methods*
- *Retain methods*

In the next five sections, we give an overview of the main problem issues related to these five areas, and exemplify how they are solved by some exist-ing methods. Our examples will be drawn from the six systems PROTOS, CHEF, CASEY, PATDEX, BOLERO, and CREEK. In the recently published book by Janet Kolodner [37] these problems are discussed and elaborated to substantial depth, and hints and guidelines on how to deal with them are given.

4. Representation of Cases

A case-based reasoner is heavily dependent on the structure and content of its collection of cases - often referred to as its case memory. Since a problem is solved by recalling a previous experience suitable for solving the new problem, the case search and matching processes need to be both effective and reasonably time efficient. Further, since the experience from a problem just solved has to be retained in some way, these requirements also apply to the method of integrating a new case into the memory. The representation problem in CBR is primarily the problem of deciding what to store in a case, finding an appropriate structure for describing case contents, and deciding how the case memory should be organized and indexed for effective retrieval and reuse. An additional problem is how to integrate the case memory structure into a model of general domain knowledge, to the extent that such knowledge is incorporated.

In the following subsection, two influential case memory models are briefly reviewed: The dynamic memory model of Schank and Kolodner, and the category-exemplar model of Porter and Bareiss[5].

[5]Other early models include Rissland and Ashley's HYPO system [52] in which cases are grouped under a set of domain-specific dimensions, and Stanfill and Waltz' MBR model, designed for parallel computation rather than knowledge-based matching.

4.1. The Dynamic Memory Model

As previously mentioned, the first system that may be referred to as a case-based reasoner was Kolodner's CYRUS system, based on Schank's dynamic memory model [54]. The case memory in this model is a hierarchical structure of what is called 'episodic memory organization packets' (E-MOPs [33,34]), also referred to as generalized episodes [38]. This model was developed from Schank's more general MOP theory. The basic idea is to organize specific cases which share similar properties under a more general structure (a generalized episode - GE). A generalized episode contains three different types of objects: *Norms*, *cases* and *indices*. Norms are features common to all cases indexed under a GE. Indices are features which discriminate between a GE's cases. An index may point to a more specific generalized episode, or directly to a case. An index is composed of two terms: An index name and an index value.

Fig. 3 illustrates this structure. The figure illustrates a complex generalized episode, with its underlying cases and more specific GE. The entire case memory is a discrimination network where a node is either a generalized episode (containing the norms), an index name, index value or a case. Each index-value pair points from a generalized episode to another generalized episode or to a case. An index value may only point to a single case or a single generalized episode. The indexing scheme is redundant, since there are multiple paths to a particular case or GE. This is illustrated in the figure by the indexing of case1.

When a new case description is given and the best matching is searched for, the input case structure is 'pushed down' the network structure, starting at the root node. The search procedure is similar for case retrieval as for case storing. When one or more features of the case matches one or more features of a GE, the case is further discriminated based on its remaining features. Eventually, the case with most features in common with the input case is found[6]. During storing of a new case,

[6]This may not be the right similarity criterion, and is mentioned just to illustrate the method. Similarity criteria may favour matching of a particular subset of features, or there may be other means of assessing case similarity. Similarity assessment criteria can in turn can be used to guide the search - for example by identifying which indexes to follow first if there is a choice to be made.

CBR研究可归纳为五个领域。一个 t
对这些问题的连贯解决方案构成了 a
案例推理方法:

- *Knowledge representation*
- *Retrieval methods*
- *Reuse methods*
- *Revise methods*
- *Retain methods*

在接下来的五个章节中，我们概述了…… e
与这五个领域相关的主要问题，并举例说明这些问
题是如何通过某些现有方法得以解决的。 克
方法。我们的示例将取自于si x
系统 PROTOS、CHEF、CASEY、PATDEX、BOL
ERO 和 CREEK。在珍妮特·科洛德纳最近出版的著
作[37]中，这些问题被 e
被深入地讨论并详尽地阐述，一个 d
关于如何处理这些问题的提示和指南 e
给定的。

4. 案例的表示

基于案例的推理系统在很大程度上依赖于其案
例库的结构与内容——该案例库通常被称为其“案
例记忆”。由于一个问题i s
通过回忆先前的、合适的经历得以解决 r
解决新问题时，案例检索与匹配 克
这些过程既需要有效，又需合理。 y
时间效率高。此外，由于刚刚解决的问题所积累的
经验必须以某种方式加以保留。 ，
这些要求同样适用于集成方法。 -
将一个新的案例写入内存。Th e
案例推理中的表示问题，主要在于确定应将哪些内
容存储于案例中，并寻找一种 n
用于描述案例内容的适当结构，以及确定案例记忆
应如何组织 d
并进行索引以实现高效检索与复用。另一个问题是
如何集成该案例 e
将记忆结构建模为通用领域知识，以至于此类知识
s
已纳入。

在下一个小节中，将简要回顾两种具有影响力
的案例记忆模型：沙克和科洛德纳的动态记忆模型
，以及…… e
波特和巴雷伊斯的类别-范例模型[5]。

4.1. The Dynamic Memory Model

如前所述，第一个系统是ma y
被称为基于案例的推理机的是科洛德纳的CYRUS
系统，该系统基于沙克的动态记忆模型。 y
模型[54]。该模型中的案例记忆是一种分层的 -
所谓“情景记忆”的心理结构 y
组织包（E-MOPs [33,34]），亦称 d
作为广义的事件[38]。该模型是 s
源自沙克更为一般的MOP理论 。
其基本思想是将特定案例进行组织， h
在更一般的结构下具有相似的性质 e
（一种泛化发作——GE）。一种泛化发作 e
包含三种不同类型的对象：*Norms*、*cases* 和
indices。范数是所有案例共有的特征。 s
在 GE 下进行索引。索引是用于区分 GE 各个案例
的特征。一个索引可以 y
指向一个更为具体的广义事件，o r
直接指向一个案例。索引由两个术语组成。 :
索引名称和索引值。

图3展示了这一结构。该图描绘了一个复杂的广
义事件，及其所包含的底层实例和更为具体的广义
事件。整个 e
案例记忆是一种判别网络，其中节点 e
要么是一个广义的事件（包含规范） ，
一个索引名称、索引值或一个案例。每个索引 -
从一个广义的事件到另一个事件的值对点 r
广义的事件或某一具体案例。一个索引值可能 y
仅指向单一案例或单一广义事件。该索引方案存在
冗余，因为从多个路径均可到达某一特定案例或广
义事件。这 s
如图中通过案例1的索引所示。

当给出一个新的病例描述时，最佳的 t
在进行匹配时，输入的案例结构为 i s
“向下推”网络结构，从……开始 e
根节点。对于案例，搜索过程与此类似。 e
检索，如同案例存储一样。当一个或多个特征 -
如果该案例的特征与某一GE的一个或多个特征相
匹配，则根据其剩余特征对该案例进行进一步区分
。最终，具有最多 t
与输入案例具有共同特征的情况被发现[6]。在存储
新案例的过程中 ，

[5]Other early models include Rissland and Ashley's HYPO system [52] in which cases are grouped under a set of domain-specific dimensions, and Stanfill and Waltz' MBR model, designed for parallel computation rather than knowledge-based matching.

[6]This may not be the right similarity criterion, and is mentioned just to illustrate the method. Similarity criteria may favour matching of a particular subset of features, or there may be other means of assessing case similarity. Similarity assessment criteria can in turn can be used to guide the search - for example by identifying which indexes to follow first if there is a choice to be made.

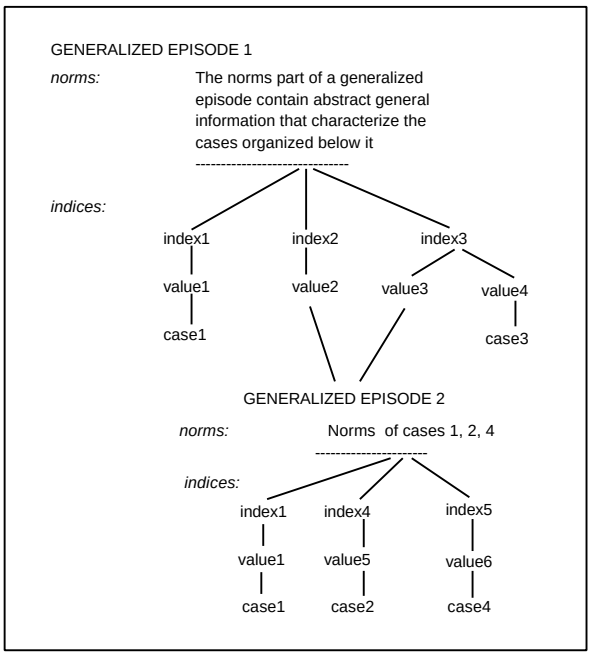


Fig. 3: Structure of cases and generalized episodes.

then discriminated by indexing them under different indices below this generalized episode. If - during the storage of a case - two cases (or two GEs) end up under the same index, a new generalized episode is automatically created. Hence, the memory structure is *dynamic* in the sense that similar parts of two case descriptions are dynamically generalized into a GE, and the cases are indexed under this GE by their difference features.

A case is retrieved by finding the GE with most norms in common with the problem description. Indices under that GE are then traversed in order to find the case which contains most of the additional problem features. Storing a new case is performed in the same way, with the additional process of dynamically creating generalized episodes, as described above. Since the index structure is a discrimination network, a case (or pointer to a case) is stored under each index that discriminates it from other cases. This may easily lead to an explosive growth of indices with increased number of cases. Most systems using this indexing scheme therefore put some limits to the choice of indices for the cases. In CYRUS, for example, only a small vocabulary of indices is permitted.

CASEY stores a large amount of information in its cases. In addition to all observed features, it retains the causal explanation for the diagnosis found, as well as the list of states in the heart failure model for which there was evidence in the patient. These states, referred to as generalized causal states, are also the primary indices to the cases.

The primary role of a generalized episode is as an indexing structure for matching and retrieval of cases. The dynamic properties of this memory organization, however, may also be viewed as an attempt to build a memory structure which integrates knowledge from specific episodes with knowledge generalized from the same episodes. It is therefore claimed that this knowledge organization structure is suitable for learning generalized knowledge as well as case specific knowledge, and that it is a plausible - although simplified - model of human reasoning and learning.

4.2. The category & exemplar model

The PROTOS system, built by Ray Bareiss and Bruce Porter [9,49], proposes an alternative way to organize cases in a case memory. Cases are also referred to as *exemplars*. The psychological and philosophical basis of this method is the view that 'real world', natural concepts should be defined extensionally. Further, different features are assigned different importances in describing a case's membership to a category. Any attempt to generalize a set of cases should - if attempted at all - be done very cautiously. This fundamental view of concept representation forms the basis for this memory model.

The case memory is embedded in a network structure of *categories*, *semantic relations*,*cases*, and *index pointers*. Each case is associated with a category. An index may point to a case or a category. The indices are of three kinds: Feature links pointing from problem descriptors (features) to cases or categories (called remindings), case links pointing from categories to its associated cases (called exemplar links), and difference links pointing from cases to the neighbour cases that only differs in one or a small number of features. A feature is, generally, described by a name and a value. A category's exemplars are sorted according

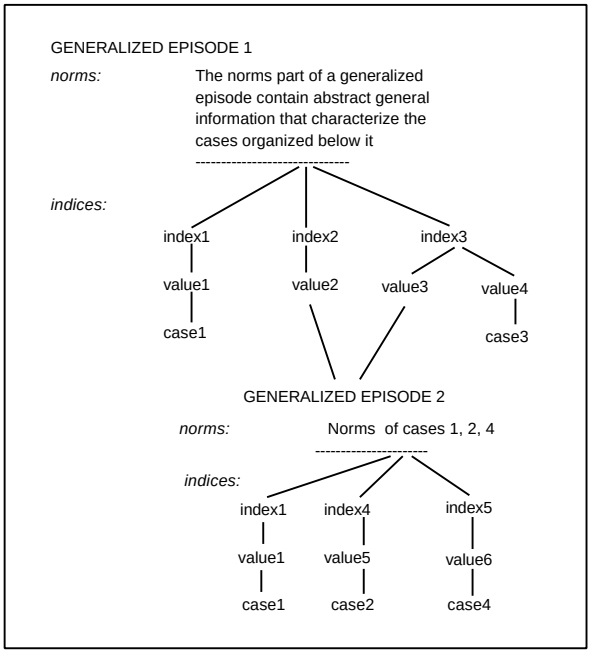


图3：案例与广义事件的结构。

然后，通过将其归入该广义事件下的不同索引进行区分。如果——在……期间 e
一个病例的存储——两个病例（或两个GE）结束u p
在相同的指标下，一个新的广义事件 i s
自动创建。因此，内存结构 i s
*dynamic*从某种意义上说，两个案例描述中的相似 ,
部分被动态地概括为一个通用等价体（GE）。
并且这些案例均按其差异在此通用等价项下进行索引。
参考特征。

通过查找具有最多匹配项的通用等价体来检索案例。
与问题描述中常见的范数 。
随后按顺序遍历该 GE 下的索引，以查找包含最多
附加信息的案例。 l
问题特征。存储新案例的过程是 i n
同样地，伴随着额外的工艺过程 f
动态地创建广义事件，a s
如上所述。由于索引结构是一种判别网络，因此一
个案例（或指向该案例的指针）i s
存储在每个索引下，以将其与其他情形区分开来。
这很可能导致爆炸性 e
指标随病例数增加而增长 。
因此，大多数采用该索引方案的系统都会对案例的
索引选择施加一定的限制。例如，在CYRUS中，仅
使用一个较小的词汇表。 f
允许使用指标。

CASEY在其病例中存储了大量信息。除了所有已观察到的特征外，它还保留了所确定诊断的因果解释，以及心力衰竭模型中存在患者相关证据的所有状态列表。这些状态被称为广义因果状态，同时也是主要的 y
将索引指向这些情况。

广义情节的主要作用是作为用于案例匹配与检索的索引结构。 。
然而，这种记忆组织的动态特性也可被视为一种尝试，旨在构建一种能够将来自特定事件的知识与从这些事件中概括出的一般性知识相整合的记忆结构 。 e
相同的事件。因此，有人认为该知识组织结构适用于 r
学习通用知识以及案例特定的知识 c
知识，并且它是一个合理——尽管是简化的——人类推理与学习的模型。

4.2. The category & exemplar model

由雷·巴雷伊斯和布鲁斯·波特构建的PROTOS系统[9,49]提出了一种替代方案。 o
在案例记忆中组织案例。案例也 o
被称为*exemplars*。该方法的心理学与哲学基础在于这样一种观点，即“rea l
在“世界”这一语境下，自然概念应当采用外延定义。此外，不同的特征被赋予 d
在描述某一案例属于某一类别时，具有不同的重要性。任何试图进行概括的尝试 a
该组案例若确实要进行，则应以验证性方式进行。y
谨慎地。这一关于概念表征的基本观点构成了该记忆模型的理论基础。

案例记忆嵌入于由*categories*、*semantic relations*,*cases*和*index pointers*构成的网络结构中。每个案例均与某一类别相关联，索引可指向某个案例或某个类别。 y。
这些索引分为三类：从问题描述符（特征）指向案例的特征链接。 r
类别（称为提醒），从类别指向其关联案例的案例链接（称为范例链接），以及从案例指向仅在某一属性上存在差异的相邻案例的差异链接。 r
少量的特征。通常而言，一个特征是 ,
由一个名称和一个值来描述。某一类别的实例按……排序

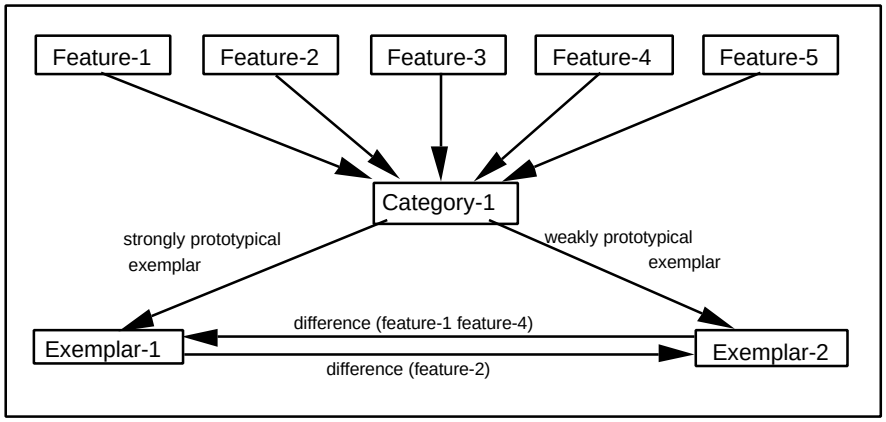


Fig. 4: The Structure of Categories, Features and Exemplars

to their degree of prototypicality in the category.

Fig. 4 illustrates a part of this memory structure, i.e. the linking of features and cases (exemplars) to categories. The unnamed indices are remindings from features to a category.

Within this memory organization, the categories are inter-linked within a semantic network, which also contains the features and intermediate states (e.g. subclasses of goal concepts) referred to by other terms. This network represents a background of general domain knowledge, which enables explanatory support to some of the CBR tasks. For example, a core mechanism of case matching is a method called 'knowledge-based pattern matching'.

Finding a case in the case base that matches an input description is done by combining the input features of a problem case into a pointer to the case or category that shares most of the features. If a reminding points directly to a category, the links to its most prototypical cases are traversed, and these cases are returned. As indicated above, general domain knowledge is used to enable matching of features that are semantically similar. A new case is stored by searching for a matching case, and by establishing the appropriate feature indices. If a case is found with only minor differences to the input case, the new case may not be retained or the two cases may be merged by following taxonomic links in the semantic network.

5. Case Retrieval

The Retrieve task starts with a (partial) problem description, and ends when a best matching previous case has been found. Its subtasks are referred to as Identify Features, Initially Match, Search, and Select, executed in that order. The identification task basically comes up with a set of relevant problem descriptors, the goal of the matching task is to return a set of cases that are sufficiently similar to the new case - given a similarity threshold of some kind, and the selection task works on this set of cases and chooses the best match (or at least a first case to try out).

While some case-based approaches retrieve a previous case largely based on superficial, *syntactical similarities* among problem descriptors (e.g. the CYRUS system [33], ARC [46], and PATDEX-1 [50] systems), some approaches attempt to retrieve cases based on features that have deeper, *semantical similarities* (e.g. the PROTOS [9], CASEY [38], GREBE [12], CREEK [3], and MMA [47] systems). Ongoing work in the FABEL project, aimed to develop a decision support system for architects, explores various methods for combined reasoning and mutual support of different knowledge types [21]. In order to match cases based on semantic similarities and relative importance of features, an extensive body of general domain knowledge is needed to produce an explanation of why two cases match and how strong the match is. Syntactic similarity assessment - sometimes referred to as a "knowledge-poor" approach - has its advantage in domains where general domain knowledge is very difficult or impossible to acquire. On the other hand, semantical oriented approaches - referred to as "knowledge-intensive"[7] - are able to use the contextual

[7]Note that syntactic oriented methods may also contain a lot of general domain knowledge, implicit in their matching methods. The distinction between knowledge-poor and knowledge-intensive is therefore related to *explicitly represented* domain knowledge. Further, it refers to *generalized* domain knowledge, since cases also contain explicit knowledge, but this

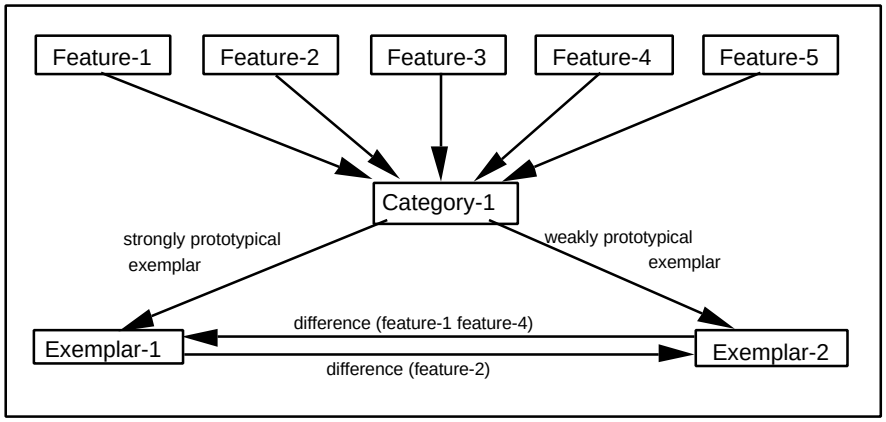


图4：类别、特征与实例的结构

在其所属范畴中的原型性程度 。

图4展示了该内存结构的一部分。 ，即，将特征与案例（实例）关联到各类别。未命名的索引是从特征指向类别的提醒信号。

在该记忆组织结构中，各类概念通过语义网络相互关联，这也 o包含特征和中间状态（例如 。由其他术语所指称的目标概念的子类 。该网络代表了一个属的背景。 l领域知识，这使得在某些基于案例的推理任务中能够提供解释性支持。例如，案例匹配的核心机制之一是一种称为“基于知识的模式匹配”的方法。

在案例库中找到与……相匹配的案例 n输入描述是通过结合输入来完成的。 t将问题案例的特征映射为指向该案例的指针。 r具有最多共同特征的类别。如果提醒 -直接指向某一类别，其链接指向该类别的最 t原型案例被遍历，且这些案例是 e已返回。如上所述，通用领域知识被用于实现特征的匹配。 t在语义上是相似的。一个新的案例被存储在b y通过搜索匹配案例，并通过建立 e适当的特征指标。如果发现一个案例具有 h如果与输入案例仅存在细微差异，新案例可能不会被保留，或者两个案例可能会被合并。 d通过遵循语义网络中的分类学链接 。

5. 案例检索

“检索”任务始于（部分）问题描述，并在找到最佳匹配的先前案例时结束。 s已发现该案例。其子任务被引用至 s识别特征、初步匹配、搜索与选择 ，按该顺序执行。识别任务基 -系统会生成一组相关的故障描述符，匹配任务的目标是返回 a在给定某种相似性阈值的情况下，选取与新案例足够相似的一组案例，并在此基础上进行选择任务。 s最佳匹配（或至少是首个可供尝试的案例）。

尽管一些基于案例的方法主要依据问题描述中的表面特征来检索先前的案例（例如，CYRUS系统[33]、ARC[46]以及PATDEX-1[50]，]系统），一些方法试图基于更深层次的特征来检索案例（例如，PROTOS [9]、CASEY [38]、GREBE [12]、CREEK [3] 和 MMA [47] 系统）。 。FABEL项目中的持续性研究工作旨在开发 p一种面向建筑师的决策支持系统，探讨了多种 -用于联合推理与互斥的多种方法 l支持不同类型的知识[21]。为了 o基于语义相似性和特征的相对重要性进行案例匹配，已有大量相关研究 l需要领域知识才能给出两个案例为何匹配以及匹配程度如何的解释。 s。句法相似性评估——有时也被称为“知识贫乏”方法——在那些通用领域知识较为匮乏的领域中具有其优势。 y难以或无法获取。另一方面 ，语义导向的方法——被称为“知识密集型”[7]——能够利用上下文

[7]Note that syntactic oriented methods may also contain a lot of general domain knowledge, implicit in their matchin克 methods. The distinction between knowledge-poor and knowledge-intensive is therefore related t*explicitly represented* domain knowledge. Further, it refers to *generalized* domain knowledge, since cases also contain explicit knowledge, but this

meaning of a problem description in its matching, for domains where general domain knowledge is available.

A question that should be asked when deciding on a retrieval strategy, is the purpose of the retrieval task. If the purpose is to retrieve a case which is to be adapted for reuse, this can be accounted for in the retrieval method. Approaches to 'retrieval for adaptation' have for example been suggested for retrieval of cases for design problem solving [15], and for analogy reasoning [17,45].

5.1. Identify Feature

To identify a problem may involve simply noticing its input descriptors, but often - and particularly for knowledge-intensive methods - a more elaborate approach is taken, in which an attempt is made to 'understand' the problem within its context. Unknown descriptors may be disregarded or requested to be explained by the user. In PROTOS, for example, if an input feature is unknown to the system, the user is asked to supply an explanation that links the feature into the existing semantic network (category structure). To understand a problem involves to filter out noisy problem descriptors, to infer other relevant problem features, to check whether the feature values make sense within the context, to generate expectations of other features, etc. Other descriptors than those given as input, may be inferred by using a general knowledge model, or by retrieving a similar problem description from the case base and use features of that case as expected features. Checking of expectations may be done within the knowledge model (cases and general knowledge), or by asking the user.

5.2. Initially Match

The task of finding a good match is typically split into two subtasks: An initial matching process which retrieves a set of plausible candidates, and a more elaborate process of selecting the best one among these. The latter is the Select task, described below. Finding a set of matching cases is done by using the problem descriptors (input features) as indexes to the case memory in a direct or indirect way. There are in principle three ways of retrieving a case or a set of cases: By following direct index pointers from problem features, by searching an index structure, or by searching in a model of general domain knowledge. PATDEX implements the first strategy for its diagnostic reasoning, and the second for test selection. A domain-dependent, but global similarity metric is used to assess similarity based on surface match. Dynamic memory based systems takes the second approach, but general domain knowledge may be used in combination with search in the discrimination network. PROTOS and CREEK combines one and three, since direct pointers are used to hypothesize a candidate set which in turn is justified as plausible matches by use of general knowledge.

Cases may be retrieved solely from input features, or also from features inferred from the input. Cases that match all input features are, of course, good candidates for matching, but - depending on the strategy - cases that match a given fraction of the problem features (input or inferred) may also be retrieved. PATDEX uses a global similarity metric, with several parameters that are set as part of the domain analysis. Some tests for relevance of a retrieved case is often executed, particularly if cases are retrieved on the basis of a subset of features. For example, a simple relevance test may be to check if a retrieved solution conforms with the expected solution type of the new problem. A way to assess the degree of similarity is needed, and several 'similarity metrics' have been proposed, based on surface similarities of problem and case features.

Similarity assessment may also be more knowledge-intensive, for example by trying to understand the problem more deeply, and using the goals, constraints, etc. from this elaboration process to guide the matching [3]. Another option is to weigh the problem descriptors according to their importance for characterizing the problem, during the learning phase. In PROTOS, for example, each feature in a stored case has assigned to it a degree of importance for the solution of the case. A similar mechanism is adopted by CREEK, which stores both the predictive strength (discriminatory value) of a feature with respect to the set of cases, as well as a feature's criticality, i.e. what influence the lack of a feature has on the case solution.

5.3. Select

From the set of similar cases, a best match is chosen. This may have been done during the initial match process, but more often a set of cases are returned from that task. The best matching case is usually determined by evaluating the degree of initial match more closely. This is done by an attempt to generate explanations to justify non-identical features,

is *specialized* (specific) domain knowledge.

问题描述在其匹配中的含义，fo r
在通用领域知识可用的领域中。

在作出决策时，应当提出的一个问题是 a
检索策略即为检索任务的目的。若检索目的在于获
取一个待经改编后复用的案例，则可在检索过程中
予以考虑。 -
检索方法。用于“适应性检索”的方法 n'
例如，已被提议用于案例检索 s
用于设计问题求解[15]，以及用于模拟电路 y
推理[17,45]。

5.1. Identify Feature

识别一个问题可能只需观察其输入描述符，但
通常——尤其是对于—— r
知识密集型方法——一种更为详尽的 e
采用了一种方法，其中试图进行…… o
在问题的上下文中“理解”该问题。未知 n
描述符可能会被忽略或要求提供。 e
由用户解释。例如，在PROTOS中，如果一个 n
输入特征对系统而言是未知的，用户 i s
要求提供一个将该特征联系起来的解释 e
融入现有的语义网络（类别结构）。理解一个问题
意味着进行过滤。 r
输出噪声问题描述符，以推断其他相关的问题特征
，并检查该特征的取值。 s
在上下文中具有合理性，以产生预期效果 s
以及其他特征等。除这些描述符之外的其他描述符e
作为输入给定，可通过使用一种通用方法进行推断。l
知识模型，或者通过从案例库中检索相似的问题描
述，并利用其特征 t
案例符合预期特征。验证预期 s
可能在知识模型内完成（案例 a） d
通用知识），或通过询问用户。

5.2. Initially Match

寻找最佳匹配的任务通常被分解为两个子任务
：一个初始匹配过程，其 h
检索出一组合理的候选方案，并且进一步 e
在众多选项中选择最佳方案的详细过程 克
这些。后者即为选择任务，如下所述。 。
通过使用……来查找一组匹配案例。 e
问题描述符（输入特征）作为索引至… e
以直接或间接的方式涉及案例记忆。存在 i n
原则：检索案例或案例集的三种方式 f
案例：通过从问题特征出发沿直接索引指针追踪，
或通过搜索索引结构，或 b y
在通用领域知识模型中进行搜索

。

is *specialized* (specific) domain knowledge。

PATDEX 实施了其诊断策略中的第一种方法。 -
时序推理，而第二个用于测试选择。基于表面匹配
，采用一种依赖于特定领域的全局相似性度量来评
估相似性。动态 c
基于内存的系统采用第二种方法，但 t
通用领域知识可被用于组合中。 n
在判别网络中进行搜索。PROTOS 和 CREEK 分别
将一和三相结合，因为它们使用直接指针来假设一
个候选集，而 h
进而通过使用……被论证为合理的匹配 f
一般知识。

病例可仅从输入特征中检索。 ，
或也来自从输入中推断出的特征。案例 s
与所有输入特征相匹配的，当然是好的。 d
匹配候选，但——取决于…… e
策略——与给定比例相匹配的案例 e
问题特征（输入或推断得到的）也可能被重新 -
检索。PATDEX采用一种全局相似性度量，具有 h
作为域的一部分而设置的若干参数 n
分析。用于评估所检索案例相关性的某些测试 s
经常执行，尤其是在病例被检索时 n
基于一组特征子集。例如，一种简单的相关性检
验可能是检查所检索到的解是否 n
与新问题的预期解类型相符。评估相似度的一种方
法是 s
必要时，已提出了若干“相似性度量” n
基于问题的表面相似性，提出了 d
案例特征。

相似性评估也可能更加依赖知识，例如通过更深
入地理解问题，并利用…… e
从这一细化过程中获取目标、约束条件等。 o
引导匹配[3]。另一种选择是加权 th e
根据其重要性对问题描述符进行 r
在学习阶段对问题进行表征 。
例如，在PROTOS中，存储案例中的每一项特征都
被赋予了一个重要性权重。 e
该案例的解决方案。CREEK 也采用了类似的机制
，它同时存储某一特征相对于……的预测强度（区
分度）。 e
一组案例，以及某一特征的临界性，即什么 t
某一特征的缺失对案例求解结果的影响 。

5.3. Select

从相似案例集中，找到最佳匹配 i s
已选定。这可能是在初始化过程中完成的。 l
匹配过程，但更常见的是一个案例集 e
从该任务中返回。最佳匹配案例 i s
通常通过评估初始程度来确定 l
更加紧密地匹配。这是通过尝试生成 -
对非相同特征的合理解释 ，

based on the knowledge in the semantic network. If a match turns out not to be strong enough, an attempt to find a better match by following difference links to closely related cases is made. This subtask is usually a more elaborate one than the retrieval task, although the distinction between retrieval and elaborate matching is not distinct in all systems. The selection process typically generate consequences and expectations from each retrieved case, and attempts to evaluate consequences and justify expectations. This may be done by using the system's own model of general domain knowledge, or by asking the user for confirmation and additional information. The cases are eventually ranked according to some metric or ranking criteria. Knowledge-intensive selection methods typically generate explanations that support this ranking process, and the case that has the strongest explanation for being similar to the new problem is chosen. Other properties of a case that are considered in some CBR systems include relative importance and discriminatory strengths of features, prototypicality of a case within its assigned class, and difference links to related cases.

6. Case Reuse

The reuse of the retrieved case solution in the context of the new case focuses on two aspects: (a) the differences among the past and the current case and (b) what part of retrieved case can be transferred to the new case. The possible subtasks of Reuse are Copy and Adapt.

6.1. Copy

In simple classification tasks the differences are abstracted away (they are considered non relevant while similarities are relevant) and the solution class of the retrieved case is transferred to the new case as its solution class. This is a trivial type of reuse. However, other systems have to take into account differences in (a) and thus the reused part (b) cannot be directly transferred to the new case but requires an *adaptation* process that takes into account those differences.

6.2. Adapt

There are two main ways to reuse past cases[8]: (1) reuse the past case solution (transformational reuse), and (2) reuse the past method that constructed the solution (derivational reuse). In transformational reuse the past case solution is not directly a solution for the new case but there exists some knowledge in the form of transformational operators {T} such that applied to the old solution they transform it into a solution for the new case. A way to organize this T operators is to index them around the differences detected among the retrieved and current cases. An example of this is CASEY, where a new causal explanation is built from the old causal explanations by rules with condition-part indexing differences and with a transformational operator T at the action part of the rule. Transformational reuse does not look how a problem is solved but focuses on the equivalence of solutions, and this is one requires a strong domain-dependent model in the form of transformational operators {T} plus a control regime to organize the operators application.

Derivational reuse looks at how the problem was solved in the retrieved case. The retrieved case holds information about the method used for solving the retrieved problem including a justification of the operators used, subgoals considered, alternatives generated, failed search paths, etc. Derivational reuse then reinstantiates the retrieved method to the new case and "replays" the old plan into the new context (usually general problem solving systems are seen here as planning systems). During the replay successful alternatives, operators, and paths will be explored first while filed paths will be avoided; new subgoals are pursued based on the old ones and old subplans can be recursively retrieved for them. An example of derivational reuse is the Analogy/Prodigy system [70] that reuses past plans guided by commonalties of goals and initial situations, and resumes a means-ends planning regime if the retrieved plan fails or is not found.

7. Case Revision

When a case solution generated by the reuse phase is not correct, an opportunity for learning from failure arises. This phase is called case revision and consists of two tasks: (1) evaluate the case solution generated

[8]We here adapt the distinction between transformational and derivational analogy, put forth in [16].

基于语义网络中的知识。如果匹配程度不够强，则尝试……
通过遵循差异链接找到更优的匹配 t
建立了密切关联的病例。该子任务通常
比检索任务更为复杂的匹配任务，尽管检索与复杂匹配之间的区别在于……
并非在所有系统中都具有显著性。选择过程通常会从每个检索到的案例中产生相应的后果与预期，并试图对这些后果进行评估并为预期提供合理性论证。这可能
通过使用系统自身的生成模型完成
领域知识，或通过向用户询问
确认及补充信息。这些病例是
最终根据某种度量或排序进行排名
标准。知识密集型选择方法
通常会生成支持这一排序过程的解释，而该案例具有最强的
对与新问题相似性的解释
已选定。此外，还考虑了该案例的其他属性。
在某些基于案例的推理系统中，特征的相对重要性与区分能力、案例在其所属类别中的原型性，以及差异链接等要素均被纳入考量。
相关案例。

6. 案例复用

在新案例的背景下，对已检索案例解决方案的重用主要集中在两个方面：(a) th
过去案例与当前案例之间的差异
(b) 可将已检索案例的哪些部分迁移到新案例中。重用的可能子任务包括复制与适应。

6.1. Copy

在简单的分类任务中，差异被抽象化处理（被视为不相关），而检索到的案例的解类则直接迁移到新案例中。
解决方案类。这是一种最简单的复用形式。然而，其他系统必须考虑(a)中的差异，因此复用的部分(b)无法直接迁移到新案例中，而需要一个能够顾及这些差异的*adaptation*过程。

6.2. Adapt

重用过往案例主要有两种方式：（1）重用过往案例的解决方案（转换式重用）。
以及 (2) 重用先前构建的 方法
解决方案（派生复用）。在变换式复用中
过去的案例解并非直接作为该问题的解。
新的案例，但存在一些以变换算子{T}形式表示的知识，这些算子作用于旧的解时，可将其转化为该新案例的解。
新案例。组织这些T算子的一种方法是t
根据检索到的案例与当前案例之间所检测到的差异，对它们进行索引。一个典型的例子是CASEY，它通过带有条件的规则，基于旧的因果解释构建新的因果解释。
部分索引差异，并通过一种变换
规则动作部分的算子 T
转换式复用并不关注问题的求解方式，而是聚焦于解的等价性。
而这一点需要一种强领域依赖的模型，其形式为转换算子{T}。
此外，还设有一个控制机制以规范操作人员的应用程序使用。

派生复用考察的是问题是如何
在检索到的案例中已得到解决。该检索到的案例包含
关于所采用的求解方法的信息
检索到的问题，包括对该操作的合理性说明
所使用的算子、所考虑的子目标、所生成的备选方案、失败的搜索路径等。推导重用
随后将检索到的方法重新实例化应用于新案例，并在新情境中“回放”旧的规划。
（通常，通用问题求解系统在此被视为
作为规划系统）。在重放过程中，成功地
将首先探索各种替代方案、操作员及路径。
而文件路径将被避开；新的子目标将被
在原有方案和子方案的基础上继续推进，且旧的子方案可递归地为它们检索。一个示例 o
派生复用是类比/天才系统[70]
该方法通过目标的共性特征复用以往的规划方案。
以及初始状态，并恢复一种手段-目的关系
如果检索到的计划失败或无效，则进入规划模式。
发现。

7. 案例修订

当通过重用阶段生成的案例解决方案时
不正确，这是从失败中学习的机会。
出现。这一阶段称为案例修订，包括
两项任务：（1）评估所生成的案例解决方案

[8]We here adapt the distinction between transformational and derivational analogy, put forth in [16].

by reuse. If successful, learn from the success (case retainment, see next section), (2) otherwise repair the case solution using domain-specific knowledge or user input.

7.1. Evaluate solution

The evaluation task takes the result from applying the solution in the real environment (asking a teacher or performing the task in the real world). This is usually a step outside the CBR system, since it - at least for a system in normal operation - involves the application of a suggested solution to the real problem. The results from applying the solution may take some time to appear, depending on the type of application. In a medical decision support system, the success or failure of a treatment may take from a few hours up to several months. The case may still be learned, and be available in the case base in the intermediate period, but it has to be marked as a non-evaluated case. A solution may also be applied to a simulation program that is able to generate a correct solution. This is used in CHEF, where a solution (i.e. a cooking recipe) is applied to an internal model assumed to be strong enough to give the necessary feedback for solution repair.

7.2. Repair fault

Case repair involves detecting the errors of the current solution and retrieving or generating explanations for them. The best example is the CHEF system, where causal knowledge is used to generate an explanation of why certain goals of the solution plan were not achieved. CHEF learns the general situations that will cause the failures using an explanation-based learning technique. This is included into a failure memory that is used in the reuse phase to predict possible shortcomings of plans. This form of learning moves detection of errors in a pot hoc fashion to the elaboration plan phase were errors can be predicted, handled and avoided. A second step of the revision phase uses the failure explanations to modify the solution in such a way that failures do not occur. For instance, the failed plan in the CHEF system is modified by a repair module that adds steps to the plan that will assure that the causes of the errors will not occur. The repair module possesses general causal knowledge and domain knowledge about how to disable or compensate causes of errors in the domain. The revised plan can then be retained directly (if the revision phase assures its correctness) or it can be evaluated and repaired again.

8. Case Retainment - Learning

This is the process of incorporating what is useful to retain from the new problem solving episode into the existing knowledge. The learning from success or failure of the proposed solution is triggered by the outcome of the evaluation and possible repair. It involves selecting which information from the case to retain, in what form to retain it, how to index the case for later retrieval from similar problems, and how to integrate the new case in the memory structure.

8.1. Extract

In CBR the case base is updated no matter how the problem was solved. If it was solved by use of a previous case, a new case may be built or the old case may be generalized to subsume the present case as well. If the problem was solved by other methods, including asking the user, an entirely new case will have to be constructed. In any case, a decision need to be made about what to use as the source of learning. Relevant problem descriptors and problem solutions are obvious candidates. But an explanation or another form of justification of why a solution is a solution to the problem may also be marked for inclusion in a new case. In CASEY and CREEK, for example, explanations are included in retained cases, and reused in later modification of the solution. CASEY uses the previous explanation structure to search for other states in the diagnostic model which explains the input data of the new case, and to look for causes of these states as answers to the new problem. This focuses and speeds up the explanation process, compared to a search in the entire domain model. The last type of structure that may be extracted for learning is the problem solving method, i.e. the strategic reasoning path, making the system suitable for derivational reuse.

Failures, i.e. information from the Revise task, may also be extracted and retained, either as separate failure cases or within total-problem cases. When a failure is encountered, the system can then get a reminding to a previous similar failure, and use the failure case to improve its understanding of - and correct - the present failure.

8.2. Index

通过复用实现。若成功，则从该成功案例中汲取经验（cess
保留（参见下一节），（2）否则，利用领域特定 e
知识修复案例解决方案，或使用 r
输入。

7.1. Evaluate solution

评估任务以应用后的结果为依据。 克
在真实环境中的解决方案（向教师请教或在现实世
界中完成任务）。这 s
通常作为案例推理系统之外的一个步骤，因为其—t—
至少对于处于正常运行状态的系统而言——涉及 e
将所提出的解决方案应用于实际问题 。
应用该解决方案后，其效果可能需要一些时间才能显现。
出现时间，具体取决于应用程序的类型 。
在医疗决策支持系统中，成功 r
治疗失败可能需要数小时至 o
数月之久。该病例仍可供学习，且b e
在中间阶段的案例库中可用 ，
但该案例必须被标记为未评估。此外，一种解决方
案也可应用于能够生成正确解的仿真程序中。这被
用于 d
在 CHEF 中，一种解决方案（即一道烹饪配方）被
应用于一个假定为强健的内部模型。 克
足以提供求解所需的必要反馈 n
修复。

7.2. Repair fault

案例修复涉及检测……的错误 e
当前解决方案以及检索或生成 克
对其的解释。最典型的例子是CHEF系统，其中因
果知识被用于生成一个 n
对解决方案规划中某些目标为何如此设定的解释 n
未能实现。CHEF能够掌握总体情况。 s
这将导致故障的发生，且基于解释的机制进行说明d
学习技术。这被纳入到一个故障 e
在重用阶段用于预测的内存 t
计划中可能存在的不足。这种学习方式将错误检测
以即席方式进行，从而 e
在细化计划阶段，错误是可以被预测的。 ，
已处理并已避免。修订的第二步 n
该阶段利用故障解释来修改…… e
以一种不会发生故障的方式进行解决方案的设计。r
例如，CHEF系统中的失败计划i s
由一个修复模块进行修改，该模块向计划中添加步
骤，以确保错误的根源将不会 t
发生。该修复模块具有通用的因果关系。 l
关于如何……的知识与领域知识 o
消除或补偿领域中的错误源。随后，修订后的计划
即可直接保留（如果…… e
修订阶段确保其正确性）或者它可以 e
再次进行评估与修复。

8. 案例保留——学习

这是将有用内容纳入的过程。 l
从新的问题解决过程中保留至 e
现有知识。从成功中汲取的经验 r
所提出的解决方案的失效是由……触发的 -
评估及可能的修复过程。它涉及 s
选择保留案例中的哪些信息，i n
以何种形式予以保留，以及如何对案件进行后期索引
从类似问题中进行检索，以及如何集成 e
内存结构中的新案例。

8.1. Extract

在基于案例的推理中，无论问题是如何解决的，
案例库都会被更新。如果问题是通过使用……来解a
决的，一个案例的基础上，可以构建一个新的案例，或者沿用
可以被推广以涵盖当前情形 s
那么，如果该问题已通过其他方法得到解决， -
包括询问用户在内，一个全新的案例将具有 e
待构建。无论如何，都需要做出决策。 e
确定应将什么用作学习的来源 。
相关问题描述符及问题解决方案 s
都是显而易见的候选者。然而，对于为何某一解能
够成为解的解释或另一种形式的论证仍然缺乏。 o
该问题也可能被标记为纳入新案例。例如，在CASE
Y和CREEK中，解释 -
这些操作被纳入保留的案例中，并在后期重复使用。s
对溶液的修改。CASEY 使用先前的 s
通过解释结构，在诊断模型中搜索能够解释新案例
输入数据的其他状态，并进一步查找这些状态的原
因。 s
对新问题的解答。这有助于聚焦并加速 s
与在整个领域模型中进行搜索相比，该方法能够加
速解释过程。最后一类结构是…… t
可用于学习的问题求解是 克
方法，即战略推理路径，使得 e
适用于派生复用的系统。

故障信息，即来自“修订”任务的信息，也可被
提取并保留，既可作为独立的故障案例，也可纳入
总体问题案例中。当…… a
当发生故障时，系统会收到先前类似故障的提醒，
并利用该 e
故障案例，以提升其对……的理解 d
正确——当前的故障。

8.2. Index

The 'indexing problem' is a central and much focused problem in case-based reasoning. It amounts to deciding what type of indexes to use for future retrieval, and how to structure the search space of indexes. Direct indexes, as previously mentioned, skips the latter step, but there is still the problem of identifying what type of indexes to use. This is actually a knowledge acquisition problem, and should be analyzed as part of the domain knowledge analysis and modeling step. A trivial solution to the problem is of course to use all input features as indices. This is the approach of syntax-based methods within instance-based and memory-based reasoning. In the memory-based method of CBR-Talk [63], for example, relevant features are determined by matching, in parallel, all cases in the case-base, and filtering out features that belong to cases with few features in common with the problem case.

In CASEY, a two-step indexing method is used. Primary index features are - as referred to in the section on representation - general causal states in the heart failure model that are part of the explanation of the case. When a new problem enters, the features are propagated in the heart failure model, and the states that explain the features are used as indices to the case memory. The observed features themselves are used as secondary features only.

8.3. Integrate

This is the final step of updating the knowledge base with new case knowledge. If no new case and index set has been constructed, it is the main step of Retain. By modifying the indexing of existing cases, CBR systems learn to become better similarity assessors. The tuning of existing indexes is an important part of CBR learning. Index strengths or importances for a particular case or solution are adjusted due to the success or failure of using the case to solve the input problem. For features that have been judged relevant for retrieving a successful case, the association with the case is strengthened, while it is weakened for features that lead to unsuccessful cases being retrieved. In this way, the index structure has a role of tuning and adapting the case memory to its use. PATDEX has a special way to learn feature relevance: A relevance matrix links possible features to the diagnosis for which they are relevant, and assign a weight to each such link. The weights are updated, based on feedback of success or failure, by a connectionist method.

In knowledge-intensive approaches to CBR, learning may also take place within the general conceptual knowledge model, for example by other machine learning methods (see next section) or through interaction with the user. Thus, with a proper interface to the user (whether a competent end user or an expert) a system may incrementally extend and refine its general knowledge model, as well as its memory of past cases, in the normal course of problem solving. This is an inherent method in the PROTOS system, for example. All general knowledge in PROTOS is assumed to be acquired in such a bottom-up interaction with a competent user.

The case just learned may finally be tested by re-entering the initial problem and see whether the system behaves as wanted.

9. Integrated approaches

Most CBR systems make use of general domain knowledge in addition to knowledge represented by cases. Representation and use of that domain knowledge involves integration of the case-based method with other methods and representations of problem solving, for instance rule-based systems or deep models like causal reasoning. The overall architecture of the CBR *system* has to determine the interactions and control regime between the CBR *method* and the other components. Note that the type of integration we address here involves case based reasoning as a core part of the target system's reasoning method, and does not include case-oriented approaches for acquiring general domain knowledge (e.g. [74]). For instance, the CASEY system integrates a model-based causal reasoning program to diagnose heart diseases. When the case-based method fails to provide a correct solution CASEY executes the model-based method to solve the problem and stores the solution as a new case for future use. Since the model-based method is complex and slow, the case-based method in CASEY is essentially a way to achieve speed-up learning. The integration of model-based reasoning is also important for the case-based method itself: the causal model of the disease of a case is what is retrieved and reused in CASEY.

An example of integrating rules and cases is the BOLERO system [40]. BOLERO is a meta-level architecture where the base-level is composed of rules embodying knowledge to diagnose the plausible pneumonias of a patient, while the meta-level is a case-based planner that, at every moment, is able to dictate which diagnoses are worthwhile to consider.

“索引问题”是一个核心且重要的问题。 h
基于案例推理中的一个核心问题。其本质在于确定
未来检索应采用何种类型的索引，以及如何构建搜
索空间。 f
索引。正如先前所述，直接索引会跳过 s
后一步骤，但仍然存在如何确定应使用何种类型索
引的问题。这i s
实际上是一个知识获取问题，并且应当 d
应作为领域知识分析与建模步骤的一部分进行分析
。针对该问题的一个简单解决方案是 s
当然，将所有输入特征用作索引。这i s
基于句法的方法在实例层面的处理方式 -
基于案例和基于记忆的推理。例如，在CBR-Talk[6
3]的基于记忆的方法中，相关 t
特征通过并行匹配来确定， 所有
案例库中的案例，并筛选出特征 t
属于与th具有较少共同特征的案例 e
问题案例。

在CASEY中，采用了一种两步索引方法。 。
主要索引特征——如在表示方法一节中所述——是
……中的通用因果状态。 e
作为解释一部分的心力衰竭模型 f
该案例。当出现新问题时，特征会在心力衰竭模型
中进行传播，并影响其状态。 s
用于解释这些特征的模型被用作案例记忆的索引。
而所观测到的特征本身则被用作…… s
仅次要特征。

8.3. Integrate

这是知识更新的最后一步。 e
在现有案例知识的基础上。如果没有新的案例，那么…
索引集已构建完成，这是Retain算法的主要步骤。
通过修改现有案例的索引 s,
案例推理系统通过学习不断提升其相似性评估能力
。现有索引的调优是案例推理学习的重要组成部分
。索引的权重 o r
对于某一特定案例或解决方案的重要性在于 e
因使用该案例的成功与否而进行调整 e
以解决输入问题。对于已被判定为对成功检索案例
具有相关性的特征，其 e
与该病例的关联性得到加强，而它i s
针对导致案例失败的特征进行了弱化处理。 s
正在被检索。由此，该索引结构具有 a
调整并使案例记忆与其使用场景相适配的作用 。
PATDEX 采用一种特殊的方法来学习特征相关性。:
相关性矩阵将可能的特征与其所对应的诊断关联起
来，并分配 a
为每条此类链路分配权重。权重会进行更新。 ,
根据成功或失败的反馈，通过一个连接 -
修复师方法。

在基于案例的推理的知识密集型方法中，
学习也可能发生在属的范围内。 l
概念知识模型，例如通过其他 r
机器学习方法（见下一节）或通过 h
与用户的交互。因此，通过一个恰当的界面 e
向用户（无论是具备相应能力的最终用户还是专家）
一个系统可能会逐步扩展并完善其功能。 s
通用知识模型及其记忆 o f
以往的案例中，在问题解决的常规过程中 。
这是PROTOS系统中的一种固有方法，用于 r
例如。PROTOS i中的所有通用知识 s
被认为是在与有能力的用户进行自下而上的交互过
程中习得的。

刚刚了解到的案例最终可能通过重新测试来验证。
输入初始问题，观察系统是否按预期运行。

9. 综合性方法

大多数案例推理系统在利用以……表示的知识的
同时，还使用了通用的领域知识。 y
案例。该领域知识的表示与应用涉及将基于案例的
方法与其他方法及表示形式进行集成。
f
问题求解，例如基于规则的系统 o r
诸如因果推理之类的深度模型。总体而言 l
CBR架构*system*必须确定CBR*method*与其他组件之
间的交互关系及控制机制。需要注意的是，我们在
此探讨的集成类型将基于案例的推理作为目标系统
推理方法的核心组成部分，而不包括用于获取通用
领域知识的面向案例的方法（例如[74]）。例如，C
ASEY系统集成了 s
一种基于模型的因果推理程序，用于诊断 e
心脏疾病。当基于案例的方法失效时， o
CASEY执行模型以提供正确的解决方案。 -
基于……的方法来解决该问题，并存储…… e
将该解决方案作为未来使用的新增案例。由于该模型
基于案例的方法复杂且耗时，案例库 d
CASEY 中的方法本质上是一种实现……的方式。 e
加速学习。基于模型的推理在案例推理方法中同样
至关重要：某一案例的疾病因果模型正是 t
在CASEY中被检索并重用。

将规则与案例相结合的一个示例是 e
BOLERO 系统 [40]。BOLERO 是一种元层级 l
一种架构，其中基础层由规则构成。 s
将知识融入以诊断其合理性 e
患者的肺炎，而元层面则是 a
基于案例的规划器，在每一时刻都能够 o
决定哪些诊断值得考虑 r。

Thus in BOLERO the rule-based level contains domain knowledge (how to deduce plausible diagnosis from patient facts) while the meta-level contains strategic knowledge (it plans, from all possible goals, which are likely to be successfully achieved). The case-based planner is therefore used to control the space searched by the rule-based level, achieving a form of speed-up learning. The control regime between the two components is interesting: the control passes to the meta-level whenever some new information is known at the base level, assuring that the system is dynamically able to generate a more appropriate strategic plan. This control regime in the meta-level architecture assures that the case-based planner is capable of *reactive behaviour*, i.e. of modifying plans reacting to situation changes. Also the clear separation of rule-based and case-based methods in two different levels of computation is important: it clarifies their distinction and their interaction.

The integration of CBR with other reasoning paradigms is closely related to the general issue of architectures for unified problem solving and learning. These approach is a current trend in machine learning with architectures such as Soar, Theo, or Prodigy. CBR as such is a form of combining problem solving (through retrieval and reuse) and learning (through retainment). However, as we have seen, other forms of representation and reasoning are usually integrated into a CBR system and thus the general issue is an important dimension into CBR research. In the CREEK architecture, the cases, heuristic rules, and deep models are integrated into a unified knowledge structure. The main role of the general knowledge is to provide explanatory support to the case-based processes [3], rules or deep models may also be used to solve problems on their own if the case-based method fails. Usually the domain knowledge used in a CBR system is acquired through knowledge acquisition in the normal way for knowledge-based systems. Another option would be to also learn that knowledge from the cases. In this situation it can be learnt in a case-based way or by induction. This line of work is currently being developed in Europe by systems like the Massive Memory Architecture and INRECA [41]. These systems are closely related to the *multistrategy learning systems* [42]: the issues of integrating different problem solving and learning methods are essential to them.

The Massive Memory Architecture (MMA) [47] is an integrated architecture for learning and problem solving based on reuse of case experiences retained in the systems memory. A goal of MMA is understanding and implementing the relationship between learning and problem solving into a reflective or introspective framework: the system is able to inspect its own past behaviour in order to learn how to change its structure so as to improve is future performance. Case-based reasoning methods are implemented by retrieval methods (to retrieve past cases), a language of preferences (to select the best case) and a form of derivational analogy (to reuse the retrieved method into the current problem). A problem in the MMA does not use one CBR method, since several CBR methods can be programmed for different subgoals by means of specific retrieval methods and domain-dependent preferences. Learning in MMA is viewed as a form of introspective inference, where the reasoning is not about a domain but about the past behaviour of the system and about ways to modify and improve this behaviour. This view supports integration of case-based learning as well as of other forms of learning from examples, like inductive methods, which are also integrated into the MMA and combined with case-based methods.

10. Example applications and tools

As a relatively young field, CBR can not yet brag about a lot of fielded applications. But there are some. We briefly summarize two of these systems, to illustrate how CBR methods can successfully realize knowledge-based decision support systems.

10.1. Two Fielded Applications

At Lockheed, Palo Alto, the first fielded CBR system was developed. The problem domain is optimization of autoclave loading for heat treatment of composite materials [26]. The autoclave is a large convection oven, where airplane parts are treated in order to get the right properties. Different material types need different heating and cooling procedures, and the task is to load the autoclave for optimized throughput, i.e. to select the parts that can be treated together, and distribute them in the oven so that their required heating profiles are taken care of. There are always more parts to be cured than the autoclave can take in one load. The knowledge needed to perform this task reasonably well used to reside in the head of a just a few experienced people. There is no theory and very few generally applicable schemes for doing this job, so to build up experience in the form of previously

因此，在BOLERO中，基于规则的层次包含 s
领域知识（如何推导出合理的诊断） 是
从患者事实中）而元层面包含 s
战略知识（它从所有可能的目标中进行规划） s,
这些很可能能够成功实现）。Th e
因此，基于案例的规划器被用于控制…… e
由基于规则的层级所搜索的空间，实现了 a
加速学习的形式。控制机制之间 n
这两个组成部分之间存在一种有趣的交互：每当出
现某些新信息时，控制权便会转移到元层面。 s
在基础层面已知，从而确保系统 i s
能够动态地生成一个更为适当 e
战略计划。这种控制机制在元堤中 l
该架构确保基于案例的规划器 i s
能够执行*reactive behaviour*，即能够根据情境变化
对计划进行调整。此外，还存在明确的分离 n
基于规则的方法与基于案例的方法在两种不同情境下的应用
计算层次的重要性在于：它阐明了它们 r
区分及其相互作用。

案例推理与其他推理范式的集成与统一问题求解与学习的体系结构这一总体性问题密切相关。 。
这种方法是当前机器学习领域的一个研究趋势。 克
诸如Soar、Theo或Prodigy等架构 。
案例推理本身是一种结合问题求解（通过检索与重用）与学习（通过……）的形式。 h
保留）。然而，正如我们所见，其他形式的… f
表示与推理通常被集成在一起。 d
转化为一个基于案例的推理系统，因此，一般性问题便是 l
将这一重要维度纳入案例推理研究。在CREEK架构中，案例、启发式规则与深度模型被整合为统一的知识结构。通用知识的主要作用是…… o
为案例库提供解释性支持 d
过程[3]、规则或深度模型也可能被使用 d
如果基于案例的方法失效，他们能够自行解决问题
。通常，在……中所使用的领域知识 a
案例推理系统是通过知识获取的。 e
以常规方式获取知识库 d
系统。另一种选择是，还可以学习…… t
从这些案例中获得的知识。在这种情况下，它可以 e
以案例为基础或通过归纳学习。这条线… f
相关工作目前正由sys在欧洲开展。 -
诸如大规模内存架构和INRECA [41]等技术。这些
系统与……密切相关 e
multistrategy learning systems[42]：整合不同问题
求解与学习方法所面临的问题 克
方法对它们至关重要。

大规模内存架构（MMA）[47]是一种基于对已存储案例经验的重用来进行学习与问题求解的集成架构。 n
系统内存。MMA 的一个目标是理解 克
以及实现学习之间的关系 克
并将问题解决转化为一种反思或内省的过程 e
框架：该系统能够对其自身的状态进行检查。 t
行为，以便学习如何改变其结构 e
以便提升未来的性能。案例库 d
推理方法由检索实现。 l
方法（用于检索过往案例），一种语言 o f
偏好（用于选择最佳情形）以及一种形式 o f
派生类比（以重用检索到的方法） o
当前问题）。MMA中的问题不会 t
采用一种案例推理方法，因为可以通过某种方式为
不同的子目标编程实现多种案例推理方法。 f
特定的检索方法和领域依赖性 t
偏好。在多智能体系统中，学习被视为一种形式的f
内省式推理，其中的推理对象并非某个领域，而是
关于过去的推理行为。 e
系统以及关于如何修改和改进该系统的途径 是
行为。这一观点支持案例的整合 -
基于学习以及其他形式的学习 克
从示例中，例如归纳法，它也 o
集成到MMA中，并与案例相结合 -
基于方法。

10. 示例应用与工具

作为一门相对年轻的学科，CBR目前尚无法……克
关于大量已部署的应用程序。但存在一些 。
我们简要概述其中两个系统，以说明案例推理方法
如何成功实现…… e
基于知识的决策支持系统。

10.1. Two Fielded Applications

在洛克希德公司的帕洛阿尔托分部，首个投入使用的CBR系统得以开发。问题领域是…… s
热处理用高压灭菌器装载优化 f
复合材料[26]。高压釜是一种大型 e
对流烘箱，用于在其中对飞机部件进行处理或 -
以获得所需的性能。不同材料类型 s
需要不同的加热和冷却程序，以及…… e
任务是为实现优化的生产率而装载高压灭菌器。 t,
即，选择可同时处理的零部件，并将其在炉内合理布置，以满足各零部件所需的加热曲线。通常情况下…… s
需要固化处理的零件数量超过了 autoclave 的承载能力。 a
一项负载。执行此任务所需的知识 k
相当普遍地被认为存在于……的头脑中 a
少数有经验的人员。缺乏理论依据和验证。 y
目前尚无适用于该任务的通用方案，s o
以先前的形式积累经验

successful and unsuccessful situations is important. The motivation for developing this application was to be able to remember the relevant earlier situations. Further, a decision support system would enable other people than the experts to do the job, and to help training new personnel. The development of the system started in 1987, and it has been in regular use since the fall 1990. The results so far are very positive. The current system handles the configuration of one loading operation in isolation, and an extended system to handle the sequencing of several loads is under testing. The development strategy of the application has been to hold a low-risk profile, and to include more advanced functionalities and solutions as experience with the system has been gained over some time.

The second application has been developed at General Dynamics, Electric Boat Division [13]. During construction of ships, a frequently re-occurring problem is the selection of the most appropriate mechanical equipment, and to fit it to its use. Most of these problems can be handled by fairly standard procedures, but some problems are harder and occur less frequently. These type of problems - referred to as "non-conformances" - also repeat over time, and because regular procedures are missing, they consume a lot of resources to get solved . General Dynamics wanted to see whether a knowledge-based decision support tool could reduce the cost of these problems. The application domain chosen was the selection and adjustment of valves for on-board pipeline systems. The development of the first system started in 1986, using a rule-based systems approach. The testing of the system on real problems initially gave positive results, but problems of brittleness and knowledge maintenance soon became apparent. In 1988 a feasibility study was made of the use of case-based reasoning methods instead of rules, and a prototype CBR system was developed. The tests gave optimistic results, and an operational system was fielded in 1990. The rule-base was taken advantage of in structuring the case knowledge and filling the initial case base. IN the fall of 1991 the system was continually used in three out of four departments involved with mechanical construction. A quantitative estimate of cost reductions has been made: The rule-based system took 5 man-years to develop, and the same for the CBR system (2 man-years of studies and experimental development and 3 man-years for the prototype and operational system). This amounts to $750.000 in total costs. In the period December 90 - September 91 20.000 non-conformances were handled. The cost reduction, compared to previous costs of manual procedures, was about 10%, which amounts to a saving of $240.000 in less than one year.

There are many any other applications in test use or more or less regular use. A rapidly growing application type is "help desk systems" [36,58], where basically case-based indexing and retrieval methods are used to retrieve cases, which then are viewed as information chunks for the user, instead of sources of knowledge for reasoning[9]. Such a system can also be a first step towards a more full-fledged CBR system.

10.2. Tools

Several commercial companies offer shells for building CBR systems. Just as for rule-based systems shells, they enable you to quickly develop applications, but at the expense of flexibility of representation, reasoning approach and learning methods. In [25] Paul Harmon reviewed four such shells: ReMind from Cognitive Systems Inc., CBR Express/ART-IM from Inference Corporation, Esteem from Esteem Software Inc., and Induce-it (later renamed to CasePower) from Inductive Solutions, Inc. The first three of these were reviewed more thoroughly by Thomas Schult for the German AI journal [56]. The example of CBR Express and ART-IM is typical, since many vendors offer CBR extensions to an existing tool. On the European scene Acknosoft in Paris offers the shell KATE-CBR as part of their CaseCraft Toolbox, Isoft, also in Paris, has a shell called ReCall. TechInno in Kaiserslautern has S3-Case, a PATDEX-derived tool that is part of their S3 environment for technical systems maintenance.

As an example of functionality, the ReMind shell offers an interactive environment for acquisition of cases, domain vocabulary, indexes and prototypes. The user may define hierarchical relations among attributes and a similarity measure based on them. Indexing is done inductively by building a decision tree and allowing the user to graphically edit the importance of attributes. Several retrieval methods are supported: (1) inductive retrieval matching the most specific prototype in a prototype hierarchy, (2) nearest neighbour retrieval, and (3) SQL-like template retrieval. Case adaptation is based on formulas that adjust values based on retrieved vs. new case

[9]A gradual transition from such a system, starting with some help desk and information filtering [74] functions, and moving to an advice-giving case-based decision support tool, is the approach taken in a system currently being specified for the Norwegian oil industry [43].

成功与失败的情形都很重要。 t。
开发此应用程序的动机是…… o
能够回忆起相关的先前情形 。
此外，一个决策支持系统将能够使其他 r
比专家更擅长完成该项工作，并有助于新人员的培训。该系统的开发始于1987年，自那时起便一直被常规使用。 e
1990年秋季。迄今为止，结果非常积极。该曲 -
租户系统负责处理一个加载的配置。 克
在孤立条件下运行，并且一个用于管理多个负载时序的扩展系统正在测试中。 。
该应用的开发策略一直是保持低风险，并纳入更多 e
随着对系统的使用经验不断积累，逐步引入了高级功能与解决方案。

第二个应用程序已在通用动力公司电力推进分部开发完成[13]。 。
在船舶建造过程中，一种频繁出现的 克
问题在于选择最合适的机械设备，并使其与使用场景相匹配。大多数… f
这些问题可以通过相当标准的 d
程序，但有些问题更为复杂且耗时。 r
较少发生。这类问题被称为 s
“不符合项”也会随时间反复出现，并且由于缺乏规范的处理流程，解决这些问题往往需要耗费大量资源。通用动力 s
想要验证基于知识的决策是否 n
辅助工具可以降低这些问题的解决成本。 。
所选择的应用领域是……的选取。 d
车载管路系统阀门的调整 。
首个系统的开发始于1986年。 ，
采用基于规则的系统方法。测试过程 f
该系统在处理实际问题时最初取得了积极效果，但存在脆弱性和知识 e
维护工作的重要性很快便显现出来。1988年，针对基于案例的……进行了可行性研究。 d
推理方法而非规则，以及一个原型 e
CBR系统已被开发。测试结果表明乐观…… c
结果表明，一套运行系统已于1990年投入应用。 。
规则库在结构化过程中得到了充分利用。 克
案例知识的积累以及初始案例库的构建。1991年秋季，该系统被持续使用。 n
涉及机械建造的四个部门中有三个。已对成本降低进行了定量估算：基于规则的系统开发耗时5人年 ，而相同的工作量则需要 e
案例推理系统（历时2人年进行研究与实验） l
开发阶段以及用于原型系统和运行系统的3个人工年。这总计为75万美元。 l
成本。在1990年12月至1991年9月期间，共处理了20,000起不合格项。其成本为 -
与以往的人工成本相比，
程序的错误率约为10%，这相当于 a
在不到一年的时间内节省了24万美元。

在测试应用中还有许多其他用途。 r
或多或少的规律性使用。快速增长的 克
应用类型为“帮助台系统”[36,58]，其中 e
基于案例的索引与检索方法 s
用于检索案例，随后对这些案例进行查看。 s
为用户提供信息块，而非源代码。 f
用于推理的知识[9]。这样的系统也可以作为迈向更为完善的案例推理系统的第一步。

10.2. Tools

多家商业公司提供外壳。 r
构建基于案例的推理系统。正如针对基于规则的系统一样 ，
外壳使您能够快速开发应用程序。 s,
但以牺牲表示的灵活性为代价 ，
推理方法与学习方法。在[25]中，Paul Harmon 回顾了四种此类外壳：来自 Cognitive Systems Inc. 的 ReMind、来自 Inference Corporation 的 CBR Express/ART-IM，以及来自 Esteem Software 的 Esteem，以及来自 Inductive Solutions, Inc. 的 Induce-it（后更名为 CasePower）。其中前三个产品由 Thomas Schult 进行了更为深入的评估。 e
德国人工智能期刊[56]。CBR Express和ART-IM的案例具有典型性，因为许多供应商 s
为现有工具提供基于案例的推理扩展。在… e
欧洲市场中，位于巴黎的Acknosoft公司提供该货架 。
KATE-CBR 作为其 CaseCraft 工具箱的一部分，Isoft 同样在巴黎，还有一款名为 ReCall 的壳程序。TechInno i
凯泽斯劳滕大学拥有S3-Case，这是一款源自PATDEX的工具，
那是其S3技术环境的一部分。 l
系统维护。

作为功能的一个示例，ReMind 书架 l
提供了一个用于案例、领域词汇、索引及原型获取的交互式环境。 。
用户可以定义其中的层次关系 -
属性及其基于这些属性的相似性度量 。
索引是通过构建决策树以归纳方式完成的。 e
并允许用户以图形化方式编辑 e
属性的重要性。几种检索方法 e
支持：(1) 与 mos 匹配的归纳式检索 t
（1）在原型层次结构中检索特定原型，（2）最近邻检索，以及（3）类SQL模板检索。案例适配基于公式，该公式根据检索到的案例与新案例之间的差异来调整数值。

[9] A gradual transition from such a system, starting with some help desk and information filtering [74] functions, and moving to an advice-giving case-based decision support tool, is the approach taken in a system currently being specified for the Norwegian oil industry [43].

differences. ReMind also has the capability of representing causal relationships using a qualitative model. The first commercial products appeared in 1991 including Help-Desk systems, technical diagnosis, classification and prediction, control and monitoring, planning, and design applications. ReMind is a trade mark of Cognitive Systems Inc. and was developed with the DARPA support.

ReCall is a CBR system trademark of ISoft, a Paris based AI company, and applications include help desk systems, fault diagnosis, bank loan analysis, control and monitoring. Retrieval methods are a combination of methods (1) and (2) in ReMind, but offer standard adaptation mechanisms such as vote and analogy, and a library of adaptation methods.

The KATE-CBR tool, named CaseWork, integrates an instance-based CBR approach within a tool for inductive learning of, and problem solving from, decision trees. The inductive and case-based methods can be used separately, or integrated into a single combined method. There are editor facilities to graphically build parts of the case/index structure, and to generate user dialogues. The tool has incorporated initial results on integration of case-based and inductive methods from the INRECA project [41].

Some academic CBR tools are freely available, e.g. by anonymous ftp, or via contacting the developers.[10]

11. Conclusions and Future trends

Summarizing the paper, we can say that case-based reasoning (CBR) puts forward a paradigmatic way to attack AI issues, namely problem solving, learning, usage of general and specific knowledge, combining different reasoning methods, etc. In particular we have seen that CBR emphasizes problem solving and learning as two sides of the same coin: problem solving uses the results of past learning episodes while problem solving provides the backbone of the experience from which learning advances. The current state of the art in Europe regarding CBR is characterized by a strong influence of the USA ideas and CBR systems, although Europe is catching up and provides a somewhat different approach to CBR, particularly in its many activities related to integration of CBR and other approaches and by its movement toward the development of application-oriented CBR systems.

The development trends of CBR *methods* can be grouped around four main topics: Integration with other learning methods, integration with other reasoning components, incorporation into massive parallel processing, and method advances by focusing on new cognitive aspects. The first trend, integration of other learning methods into CBR, forms part of the current trend in ML research toward *multistrategy learning* systems. This research aims at achieving an integration of different learning methods (for instance case-based learning and induction as is done in the MMA and INRECA systems) into a coherent framework, where each learning method fulfills a specific and distinct role in the system. The second trend, integration of several reasoning methods aims at using the different sources of knowledge in a more thorough, principled way, like what is done in the CASEY system with the use of causal knowledge. This trend emphasizes the increasing importance of knowledge acquisition issues and techniques in the development of knowledge-intensive CBR systems, and the European Workshop on CBR showed a strong European commitment towards the utilization of knowledge level modeling in CBR systems design.

The massive memory parallelism trend applies case-based reasoning to domains suitable for shallow, instance-based retrieval methods on a very large amount of data. This direction may also benefit from integration with neural network methods, as several Japanese projects currently are investigating [32]. By the fourth trend, method advances from focusing on the cognitive aspects, what we particularly have in mind is the follow-up of work initiated on creativity (e.g. [55]) as a new focus for CBR methods. It is not just an 'application type', but a way to view CBR in general, which may have significant impact on our methods.

The trends of CBR *applications* are clearly that we initially will see a lot of help desk applications around. This type of systems may open up for a more general coupling of CBR - and AI in general - to information systems. The use of cases for human browsing and decision making, is also likely to lead to increased interest in intelligent computer-aided learning, training, and teaching. The strong role of user interaction, of flexible user control, and the drive towards total interactiveness of systems (of 'situatedness', if you like) favours a case-based approach to intelligent computer assistance, since CBR systems are able to continually learn from, and evolve through, the capturing and retainment of past

[10] Protos, for example, is available from the University of Texas, and code for implementing a simple version of dynamic memory, as described in [51], is available from the Institute of Learning Sciences at Northwestern University.

差异。ReMind还具备
使用定性方法表示因果关系
模型。首批商业化产品于1991年问世，其中包括帮助台系统、技术诊断、分类与预测以及控制等。
监测、规划与设计应用。ReMind 是 Cognitive Systems Inc. 的注册商标，并且……
在DARPA的支持下开发。

ReCall 是 ISoft 公司的案例推理系统商标，位于帕里。基于人工智能的公司，其应用包括帮助台系统、故障诊断、银行贷款分析、控制等。
以及监测。检索方法是 ReMind 中方法 (1) 和方法 (2) 的组合，但提供了诸如投票和类比等标准化的适应机制，一个
一套适应性方法库。

KATE-CBR工具，名为CaseWork，集成了
一种基于实例的案例推理方法，集成于一个工具之中
归纳学习与问题求解
，
决策树。归纳法与基于案例的方法
可单独使用，也可集成到单一系统中。
组合方法。系统提供了用于以图形化方式构建案例/索引结构各部分的编辑工具，一个
以生成用户对话。该工具已集成
基于案例的集成初步结果
INRECA项目中的归纳方法[41]。

一些学术性的基于案例的推理工具是免费提供的，例如 通过匿名 FTP，或直接联系开发者。[10]

11. 结论与未来发展趋势

综上所述，我们可以认为基于案例的
基于案例的推理（CBR）提出了一种范式化的方法，到
攻击性人工智能问题，即问题求解、学习
，
通用知识与专用知识的结合使用
克
不同的推理方法等。特别是，我们已经
已经看到，案例推理强调问题求解。
学习如同一枚硬币的两面：问题解决利用以往学习过程中的成果，而
问题求解构成了……的主干
从中获得学习进展的经验。当前的
欧洲在基于案例推理方面的最新研究进展
以美国理念的强烈影响为特征
以及案例推理系统，尽管欧洲正在迎头赶上。
提供了一种与案例推理略有不同的方法
，
尤其是在其与集成相关的诸多活动中
基于案例推理及其他方法，并通过其运动
面向应用型案例推理的发展
系统。

CBR v7 的发展趋势可归纳为四个主要方面：集成与
其他学习方法、与其他推理组件的集成、以及在大规模系统中的整合
并行处理，以及通过聚焦实现的方法进展
关于新的认知方面。第一种趋势，即把其他学习方法整合到案例推理中，构成了其一部分……
当前机器学习研究的主流趋势是向*multistrategy learning*系统发展。本研究旨在实现不同学习方法的集成（例如
基于案例的学习与归纳，正如在……中所做的那样
将MMA和INRECA系统整合为一个连贯的框架，
其中每种学习方法均能实现
在系统中发挥特定且独特的作用。第二种趋势，即多种推理方法的集成，旨在……
以更加全面、基于原则的方式利用不同的知识来源
，正如在……中所做的那样。
基于因果知识的CASEY系统
。
这一趋势凸显了……日益增长的重要性
知识获取中的问题与技术
知识密集型案例推理系统的开发
，
并且，欧洲基于案例推理的研讨会显示出强烈的
欧洲在基于案例推理系统设计中对知识层次建模的重视与投入。

大规模内存并行化趋势将基于案例的推理应用于适合浅层推理的领域。
，
基于实例的检索方法在非常大
数据量。这一研究方向也有望通过与神经网络方法的融合而获益，正如目前日本的多项研究项目正在探索的那样[32]。在第四种趋势中，方法的进步已从关注
在认知层面，我们尤其关注的是对创造力相关研究的后续跟进。
（例如 [55]）作为CBR方法的一个新研究方向。这并非
而不仅仅是一种“应用类型”，更是一种看待CBR的方式。总体而言，这可能会对我们的研究产生重大影响。
方法。

CBR *applications* 的发展趋势明确表明，我们最初将看到大量帮助台应用程序。
周围。这类系统可能为进一步的研究开辟新的途径。
CBR与人工智能在总体上的通用耦合
形成系统。将案例用于人类的浏览与决策，也很可能会导致……
对智能计算机辅助学习、培训与教学的兴趣日益增长。其重要作用在于
用户交互、灵活的用户控制以及驱动
朝着系统的完全交互性（
“情境性”（如果您愿意这样称呼的话）有利于基于案例的一种面向智能计算机辅助的途径，因为案例推理系统能够通过捕获与保留过往经验而不断学习并演化
。

[10] Protos, for example, is available from the University of Texas, and code for implementing a simple version of dynamic memory, as described in [51], is available from the Institute of Learning Sciences at Northwestern University.

experiences.

Case-based reasoning has blown a fresh wind and a well justified degree of optimism into AI in general and knowledge based decision support systems in particular. The growing amount of ongoing CBR research - within an AI community that has learned from its previous experiences - has the potential of leading to significant breakthroughs of AI methods and applications.

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经验。

基于案例的推理为……注入了新的活力，并且 a
对人工智能总体而言，抱有充分合理的乐观态度 d
尤其是基于知识的决策支持系统。不断增长的持续
性CB R
研究——在一个已经学习过的AI社区中 d
基于其以往的经验，有可能推动人工智能方法与应
用取得重大突破。

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